

ASSIGNMENTS IN **CHEMISTRY** BOOK 1

Peter Gribben & Wendy St Germain



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CHEMISTRY
BOOK 1

Peter Gribben & Wendy St Germain



Wendy St Germain undertook and completed her science training at Sydney's Macquarie University, majoring in zoology and genetics and graduating with distinctions in several courses. Her ability to write science clearly and in ways which clarify rather than confuse students has made her a popular author. Wendy has written many titles, both independently and as a co-author with Peter Gribben. The subjects of her books include genetics, biology, physics, chemistry, space science, aboriginal studies, english and more. She has written numerous other educational materials including educational CD ROMs, integrated units, 'in class' tests and sections of the NSW Independent Trial Examinations. Wendy has recently developed a whole new following with her children's fiction and young adult titles.

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INTRODUCTION

This series of chemistry assignments has been written by a highly-experienced team of authors. Each worksheet has been specially designed to meet the needs of busy teachers of chemistry – they require minimal preparation and do not compromise on quality of content. Most of the material requires very little prior knowledge from the student. New chemical terms are explained as they are met in the text and important concepts are revised or summarised. Areas that students have traditionally found difficult have been broken down into sections and are examined one step at a time.

There are many worked examples of calculations and sufficient questions for students to practise and master difficult concepts. Research questions have been included where appropriate to give students the opportunity to explore the history of chemistry.

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Separating solids of different sizes

When a solid mixture is made up of particles of different sizes, a sieve can be used to separate the mixture.

- Figure 1 shows two stages in the process of using a sieve to separate a solid mixture. Explain the principle behind this method of separation.
- Give an example of a mixture that could be separated using this method.

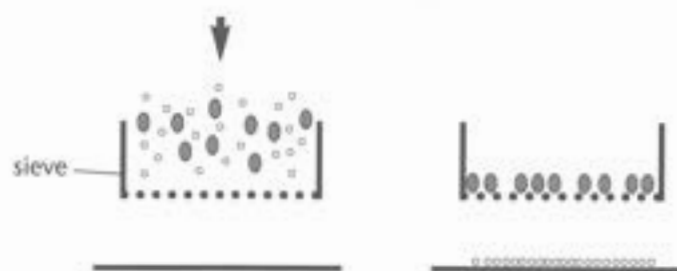


Figure 1

Separating an insoluble solid from a liquid

Filter paper is porous; it has tiny holes in it. The holes (pores) allow small particles to pass through, but hold back larger particles. Water and solutions will pass through filter paper but undissolved particles will not.

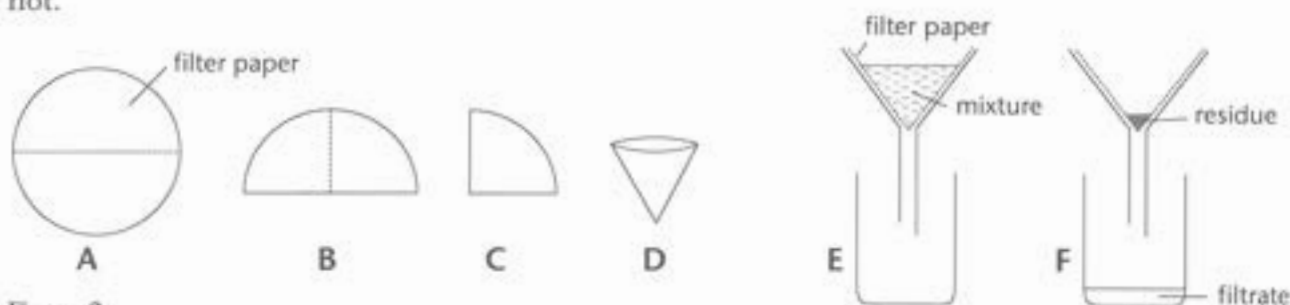


Figure 2

- Diagrams A to F in figure 2 show a sequence for filtering a mixture of water and an insoluble solid. Refer to these diagrams to help you describe how a typical filtration could be conducted.
- Explain what is meant by 'residue' and 'filtrate' as labelled in diagram F.

Separating dissolved solids from liquids

A solid can be recovered from a solution through evaporation (vaporising the liquid). As the mixture is heated, the solvent (liquid) changes to a vapour and evaporates. This leaves the solute (solid).

In a science laboratory, this process is sometimes done in two stages. Direct heat is used to concentrate the solution and then a water bath is used to complete the evaporation. A water bath consists of water being boiled in a beaker. As the water boils, the steam (at 100°C) warms the evaporating basin and its contents. A water bath provides a lower temperature than direct heat from a Bunsen burner.

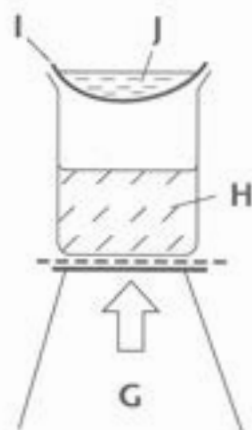


Figure 3

- Figure 3 shows a water bath being used to evaporate a solution. Identify parts G, H, I and J.
- Why does using a water bath to evaporate a solution take longer than using direct heat from a Bunsen burner?
- Evaporating basins can be made of glass or ceramic materials. Give two desirable properties of a material used to make an evaporating basin.
- Define *solute*.
- Define *vapour*.

Separating a soluble solid from an insoluble solid

10 The following instructions relate to separating a mixture of two solids. One solid is soluble and the other solid is insoluble. The instructions are out of sequence. Rewrite them in the correct sequence.

- Heat and stir if the soluble substance is not completely dissolved.
- Evaporate the filtrate to recover the soluble solid.
- Add a solvent and stir.
- Dry the residue on filter paper to recover the insoluble solid.
- Filter and then place the filtrate in an evaporating basin.
- Put the mixture into a beaker.

11 A student was given a mixture of common salt (sodium chloride) and sand. She separated these two solids using the method outlined in Question 1. Her results are shown below.

- Which solid was soluble in water?
- Which solid was insoluble in water?
- What was the mass of sand in the mixture?
- What was the mass of salt in the mixture?
- Calculate the relative percentages of sand and salt in the mixture.
- Not all of the information in the results was necessary for calculating the amounts of sand and salt present. Name one piece of data that was unnecessary.
- What property of filter paper allows it to separate an insoluble solid from a liquid?

Mass of mixture:	8.3 g
Volume of water added:	50 mL
Mass of dry filter paper:	0.1 g
Mass of filter paper plus sand (after drying):	4.9 g
Mass of evaporating basin (empty):	121.6 g
Mass of evaporating basin plus salt:	125.1 g

The centrifuge

A **centrifuge** is a device that can separate suspensions. Centrifuges spin at high speeds (up to hundreds of thousands of revolutions per minute – rpm). Forces 10 000 times that of gravity can be obtained in a centrifuge.

A domestic spin drier is a type of centrifuge. When the machine spins, excess water is flung to the outside, off the wet clothes. At the same time warm air blows in to evaporate the water.

A **suspension** is a liquid that has small particles (liquid or solid) suspended in it. For separation, samples of a suspension are placed in centrifuge tubes. These tubes resemble test tubes but can withstand much greater forces.

When the tubes are spun, the more dense particles move to the bottom of the tube. The Figure 4 shows a 'before and after' version of a mixture that has been put through a centrifuge. Centrifuges are especially important in separating biochemical mixtures such as blood.

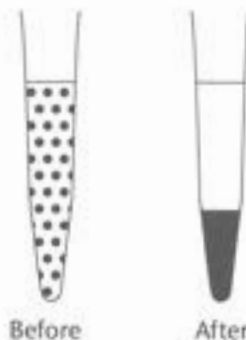


Figure 4

- Define *suspension*.
- What is meant by *rpm*?
- Outline how a centrifuge can separate a suspension.
- Name a household appliance that uses a centrifuge.
- Explain what is meant by *more dense*.
- Is a blender a type of centrifuge?

Research 18 Explain why a centrifuge is important in blood preservation.

19 Find out the components in blood that can be separated using a centrifuge.

THE THREE STATES OF MATTER

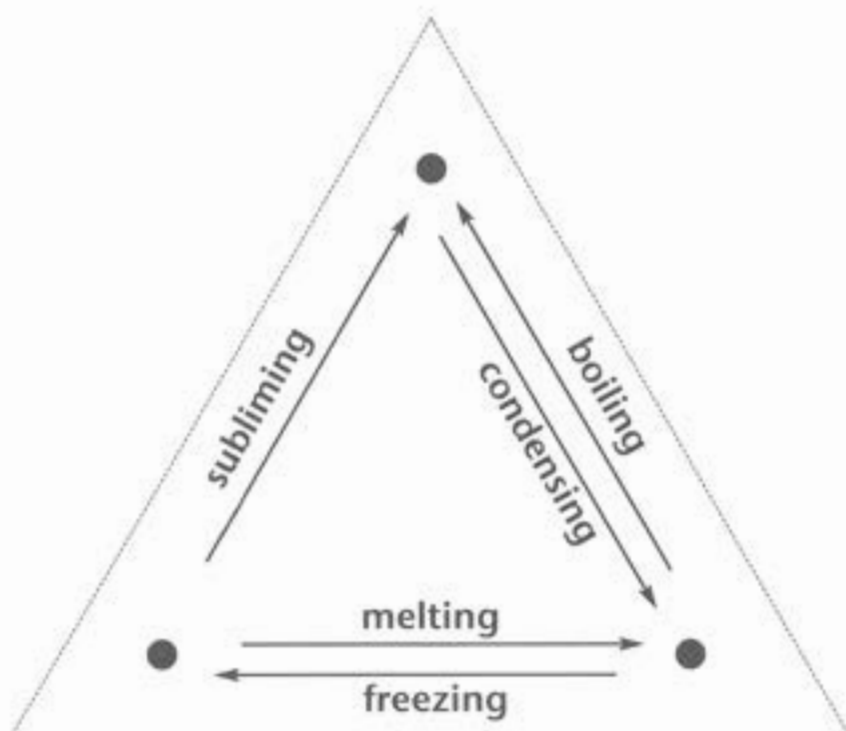
2

Matter is anything that has mass and takes up space. Matter can exist in three states or phases: solids, liquids and gases.

- 1 Cut out the headings and descriptions below. Arrange each description under the appropriate heading and then glue them in place.

Solids	Liquids	Gases
Have no fixed shape; take the shape of their containers	Have no fixed shape; take the shape of their container	Have no fixed volume; occupy the volume of their container
Have a fixed shape	Are much less dense than solids or liquids	Are difficult to compress and have a fixed volume
Are usually the most dense state	Are almost as dense as their solid phase	Have a fixed volume

- 2 Matter can change its state. The triangle below shows changes of state. Cut out the triangle and the labels. Put each label at the correct corner of the triangle. Glue the labels in place.



Solid
Liquid
Gas

The gases that surround Earth are known as its atmosphere. The atmosphere is a source of many important gases. These gases can be obtained from the air by **fractional distillation**. In this process, air is compressed and cooled until it is a liquid. The temperature is gradually increased and gases are distilled off in the order of their boiling points.

The gas with the lowest boiling point will be the first to vaporise and be distilled off. As the temperature continues to rise, the gas with the next lowest boiling point will vaporise, and so on. The gas with the highest boiling point will be the last to be distilled. Figure 1 shows a simplified version of how two gases separate during this process.

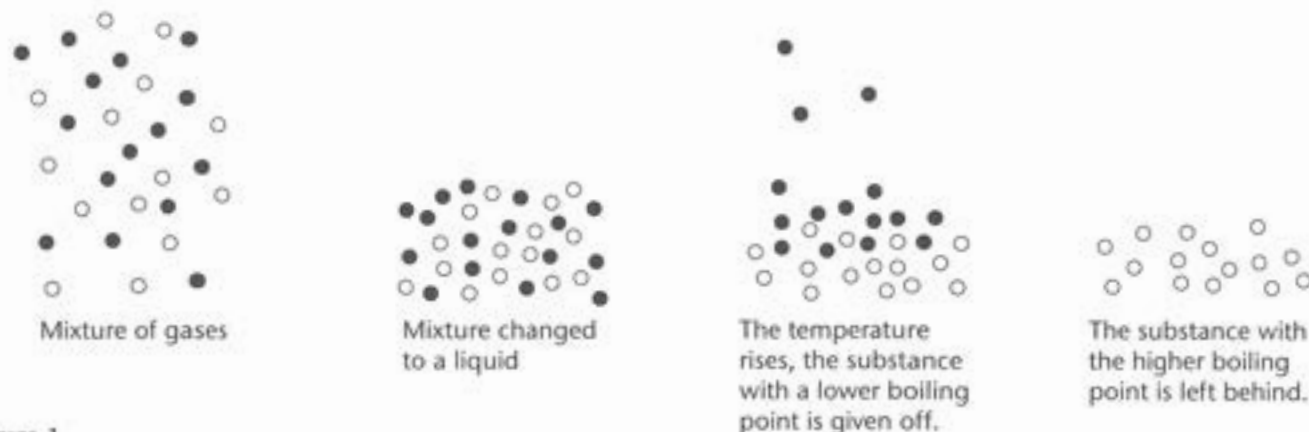


Figure 1

- 1 Define *boiling point*.

Liquid nitrogen has a lower boiling point than liquid oxygen. This means that in fractional distillation it distils off before oxygen does. Liquid nitrogen is very cold (-196°C) and can be used to freeze food very quickly (flash-freezing) or to keep food frozen during transportation. Nitrogen is a colourless, odourless, tasteless, non-toxic gas. It is very unreactive and is sometimes used to fill space in food packages such as potato chips to prevent oxidation and physical damage. It is also used in oil tanks to reduce the risk of fire.

- 2 Why is food frozen? Name one item other than food that could be flash-frozen.
- 3 Explain what is meant by *non-toxic*.
- 4 Define *cryogenic* and *superconductivity*.

Oxygen is also obtained from liquid air by fractional distillation. Oxygen is a clear, colourless gas with no odour. It is slightly soluble in water (i.e. small amounts can dissolve in water). Unlike nitrogen, it is very reactive and combines with many substances. Oxygen is compressed and put into cylinders for use by people who go where there is little or no air; e.g. people in high-altitude aircraft, firefighters, astronauts and scuba divers. It is also used in medical situations where people have breathing difficulties; e.g. oxygen tents are placed around people so that they can breathe in pure oxygen.

The '**noble gases**' (helium, neon, argon, krypton and xenon) can also be extracted from the air. They are all very unreactive substances. Helium (commercially extracted from methane) is used for lighter-than-air balloons (blimps/dirigibles) and mixed with oxygen to make a breathing mixture for scuba divers. Helium occurs as a liquid just above absolute zero (-273°C) and has a boiling point of -269°C . It can be used to cool substances close to absolute zero and is used in experiments in superconductivity and cryogenics (the study and use of substances at very low temperatures).

Neon can also be used as a cryogenic refrigerant but is more commonly applied in 'neon lighting', where electric discharge tubes used in advertising signs give a crimson colour.

Argon is used to fill incandescent filament light globes. It prevents the tungsten filament from burning. It also gives a blue light in neon lighting and is used in electric welding to prevent the metal being welded from burning. Krypton gives a bright orange-red emission in neon lighting.

- 5 Draw up a table giving the names, symbols, properties and uses of the gases mentioned on this page.

When a crystal of copper (II) sulfate is placed in a beaker of water, over a period of time the water will turn blue and the crystal will disappear.

- 1 Complete the sentence: *The copper sulfate has _____ in the water to form a _____.*

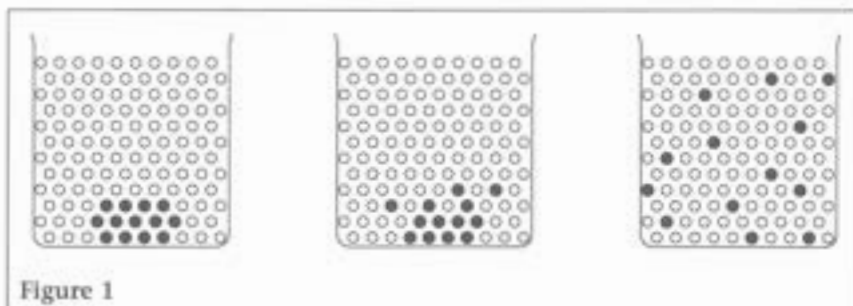


Figure 1

- 2 The crystal (solid) and the water (liquid) are both made up of particles. Use the diagrams in figure 1 to explain why the water turns blue and the crystal disappears.
- 3 Can you see the individual copper sulfate particles after the crystal has dissolved? Explain your answer.

Diffusion is the movement of particles among one another so that they become evenly mixed.

Bromine (Br) forms a red-brown vapour and is denser than air.

- 4 What colour is air? _____

When a gas jar of air is placed over a jar of bromine vapour so that the gases can mix together, the different gases look the same after a few minutes. The colour of the gases is paler than the colour of the original bromine gas. This is shown in figure 2.

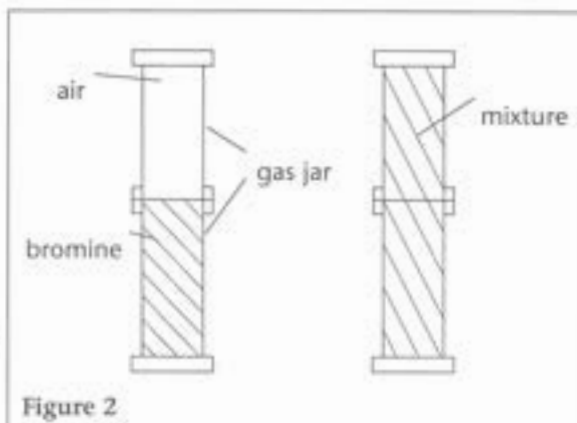


Figure 2

- 5 Complete the sentences:

Air and bromine vapour are both made up of tiny moving _____. They bump into each other and move in all directions so that they eventually become evenly mixed. This is an example of _____.

- 6 Why is the mixture of gases paler in colour than the original bromine gas?

- 7 A teacher placed a few millilitres of perfume on a glass dish at the front of the class. Then she asked the students to raise their hands when they could smell the perfume. Describe what would happen and explain why.

- 8 Generally speaking, diffusion takes place more quickly with gases than with liquids. Give a reason for this.

- 9 Kinetic energy is the energy of motion. True or false?

Brownian motion

Robert Brown was a Scottish botanist on Matthew Flinders' expedition to Australia (1801–05). On this expedition, Brown collected around 1700 new plant species. In 1827, he was examining pollen grains under a microscope. The pollen grains were suspended in water and he observed that the tiny particles were moving in a continuous, erratic manner. His first explanation was that pollen grains, being from a living organism, were moving of their own accord. This theory was disproved when a similar effect resulted from suspending in water small inorganic particles such as toothpaste and graphite (carbon).

Water molecules (which are too small to be seen) have **kinetic energy** (i.e. they are moving). These molecules hit the suspended particles at irregular intervals and from irregular directions.

The collisions between the water molecules and the particles make the particles move. This erratic motion of the visible particles is sometimes called 'the drunkard's walk' (figure 3).

For this effect to be visible, particle size is very important. The particles have to be small enough to be deflected by water molecules but large enough to be seen under a microscope. It is easier to use smoke trapped in a small glass box and examine it under a microscope.

Light is focused onto smoke by a lens (figure 4). The smoke cell is sealed so that draughts do not affect the smoke. The smoke particles are constantly bombarded by the invisible air molecules, causing a random, 'jittery' motion.

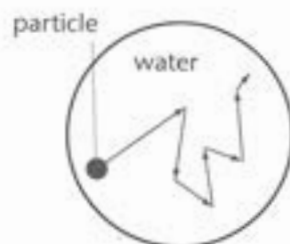


Figure 3

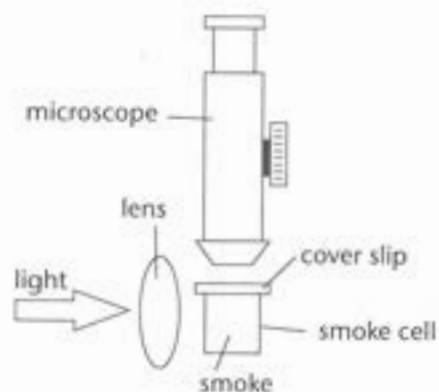


Figure 4

10 For the apparatus shown in figure 4, state the function of the:

- a smoke cell and cover slip _____
- b lens _____
- c microscope _____
- d light _____

11 Give a brief definition for each of the following terms.

- a botanist _____
- b inorganic _____
- c fluid _____
- d random _____
- e kinetic _____
- f particle _____
- g vapour _____

12 What does the following define? _____

The movement of particles from an area of greater concentration to an area of lesser concentration - spreading out equally to fill all the space available.

13 Particles have to be small enough to be deflected by water molecules. Explain this statement in regard to Brownian motion.

14 Give your own definition for *Brownian motion*.

15 Circle the correct word in brackets to complete the sentence: *Brownian motion provides evidence that the (inorganic, random, kinetic) model is accurate.*

Elements are pure substances that contain only one type of atom. Elements can be metals, non-metals or semi-metals (metalloids).

- 1 Name two metal elements. _____
- 2 Name two non-metal elements. _____

Coins are usually made of metals. Silver (Ag), copper (Cu) and gold (Au) have been used to make coins for thousands of years. These metals are also used to make jewellery.

- 3 Give two reasons why jewellery is made from silver, copper and gold.

- 4 Give two reasons why coins are made from metal.

Most metals are lustrous (shiny) and they are often silver in colour. Metals are good conductors of heat and electricity, whereas most non-metals are poor conductors.

- 5 Name one metal that is not silver in colour. _____

Metals are malleable (can be beaten into shape). They are also ductile (can be drawn out into wires). Non-metals are brittle (they snap instead of bend). Metals are usually denser (more matter is compressed into a space) than non-metals. Metals are solids at room temperature, with the exception of mercury (Hg). They usually have high melting points. Metals are found on the left side and centre of the Periodic Table.

- 6 Where in the Periodic Table do you expect non-metals to be located? _____
- 7 Give another name for *metalloids*. _____
- 8 Name two metals that are drawn into wires. _____
- 9 'Most metals are liquids at room temperature.' True or false? _____
- 10 'Elements contain only one type of atom.' True or false? _____
- 11 'Non-metals are usually denser than metals.' True or false? _____
- 12 'Semi-metals have properties of both metals and non-metals.' True or false? _____
- 13 Name each of the following elements. State whether each element is a metal or a non-metal.

- | | |
|------------|------------|
| a Al _____ | j Ar _____ |
| b B _____ | k Be _____ |
| c C _____ | l Ca _____ |
| d Cl _____ | m Cu _____ |
| e F _____ | n Fe _____ |
| f H _____ | o He _____ |
| g N _____ | p Na _____ |
| h Ne _____ | q Ni _____ |
| i S _____ | r Si _____ |

- 14 Which metal is the only metal that is a liquid at room temperature? _____

Some metals, such as iron (Fe), cobalt (Co) and nickel (Ni), can be attracted by magnets or used to make magnets. Metals make up about 25% by mass of the Earth's crust (includes oceans and atmosphere).

15 For each statement, choose and write the correct word from the brackets.

- a Something that is shiny is said to be (lustrous/dull) _____.
- b A pure substance that contains only one type of atom: (compound/element) _____.
- c Can be beaten into shape: (malleable/solid) _____.
- d Can attract another metal: (electric/magnetic) _____.
- e Can be drawn into wires: (ductile/liquid) _____.
- f Snaps when bent: (conduct/brittle) _____.
- g Allows energy such as heat to pass through easily: (conducts/melts) _____.

Sometimes elements chemically join with each other to form **compounds**. These compounds have different properties to the elements from which they were formed.

Often metals combine with non-metals. The name of the compound formed comes from the elements from which it is made. If a compound only contains two elements, its name ends in '-ide'. In a compound that contains a metal and a non-metal, the metal is first in the compound's name. For example, iron oxide (FeO) contains the metal iron (Fe) and the non-metal oxygen (O); potassium fluoride (KF) contains the metal potassium (K) and the non-metal fluorine (F).

16 For each compound, give the names and chemical symbols for the elements present.

- a copper chloride _____
- b iron oxide _____
- c sodium chloride _____
- d lead iodide _____
- e magnesium oxide _____
- f barium sulfide _____
- g calcium carbide _____
- h hydrogen bromide _____

17 Name one chemical compound not mentioned on this page. _____

18 Give the names and chemical symbols for the elements present in your answer to Q17.

There can be more than two elements present in a compound. If one of these elements is oxygen, the compound's name usually ends in '-ate'. For example, zinc sulfate (ZnSO_4) contains zinc (Zn), sulfur (S) and oxygen (O).

19 For each compound, give the names and chemical symbols for the elements present.

- a calcium sulfate _____
- b copper carbonate _____
- c silver nitrate _____
- d sodium phosphate _____
- e nickel nitrate _____
- f zinc sulphate _____
- g lead carbonate _____

Elements have been discovered at different times and in different ways. There are many different reasons for the names they are given.

Complete the table below. The first element has been done for you.

Element	Symbol	When and where discovered	Origin of name/symbol
Aluminium	Al	1826 Germany	From the Latin word for alum, ' <i>alumen</i> '. (Alum is an aluminium compound.)
	Ar		
		1774 Sweden	From the Greek word ' <i>chloros</i> ' meaning 'green light'.
Helium			
	Hg		
		Ancient	From the Latin word ' <i>sulphurium</i> '.
	U		
Oxygen			
			From the Latin name for Cyprus, ' <i>cuprum</i> ', where this metal was first produced. (Greek = ' <i>Kypros</i> ')
	C		

Lavoisier took the first step in classifying elements in 1789. He divided the twenty-five or thirty elements known to him into metals and non-metals. In 1829 Johann Döbereiner noticed that where groups of three similar elements (triads) occurred, the relative atomic mass of the middle element came approximately between those of the other two. However, for many elements the idea did not work. In 1864 John Newlands found that if the elements known to science at the time were arranged in order of increasing relative atomic masses, a pattern emerged. Starting at any element, every eighth element was 'a kind of repetition of the first'. Because of the similarity of this pattern to the musical scale, he called it the **law of octaves**. This 'law' is not very reliable.

The Periodic Table is a list of the elements arranged in order of increasing atomic number.

- 1 Explain the difference between *atomic mass* and *atomic number*.

In 1869 Dmitri Mendelée (or Mendeléeff) made a breakthrough. He arranged the elements in order of increasing relative atomic masses in a similar way to Newlands but, in order to obtain a better repetition, he left gaps for elements which he assumed must exist but had not yet been discovered. He found that many properties then followed a repeating pattern, or **periodic variation**.

Not only did Mendelée have to leave gaps for undiscovered elements, he also had to correct some relative atomic masses that had been determined erroneously to make the elements fit into the right groups. Mendelée predicted the existences of 21 missing elements including 'eka-silicon' which is now named germanium. His predicted chemical and physical properties were very close to those actually found when germanium was discovered.

The Periodic Table we now use lists the elements in order of atomic number (not atomic mass). Figure 1 shows an outline of the Periodic Table. There are just over 90 naturally occurring elements.

Chemical periodicity refers to the regular occurrence of a set of properties of an element or its compounds in the Periodic Table. Groups are vertical (columns) and are labelled in Roman numerals.

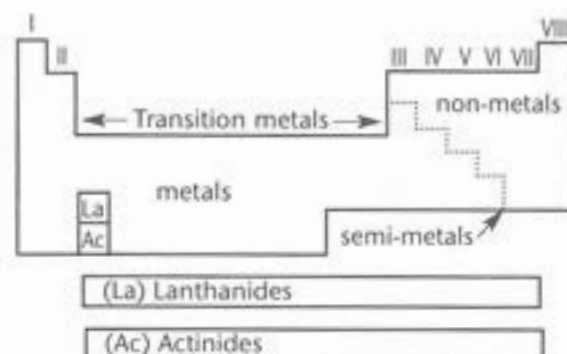


Figure 1

- 2 Draw up a table showing the Roman numerals from one to eight and their current equivalents.
- 3 Give two differences between a Periodic Table from Mendelée's time and a current Periodic Table.

Groups often contain elements with similar properties. **Periods** are horizontal. The Periodic Table can be divided into metals and non-metals. Metallic character is associated with atoms whose nuclei have a low degree of electronic control (i.e. lose electrons in bonding) whereas non-metallic character is associated with atoms whose nuclei have a high degree of control of outer-shell electrons (i.e. gain electrons in bonding).

Some elements exhibit characteristics of both metals and non-metals. These 'semi-metals' or 'metalloids' lie on either side of the dividing line (see the dotted line in Figure 1). Germanium (Ge) and silicon (Si) are examples of these elements.

- 4 Outline the contributions made by Lavoisier, Döbereiner, Newlands and Mendelée in establishing the modern Periodic Table.
- 5 Describe the differences between groups and periods in the Periodic Table.
- 6 Where in the Periodic Table are the transition metals located?
- 7 Where in the Periodic Table are the semi-metals located?
- 8 Outline the properties of semi-metals.
- 9 Give the names and symbols of two semi-metals.
- 10 Study a full Periodic Table. What information is given about each element?

Hydrogen is sometimes placed at the top of Group I metals, but the only thing it has in common with these metals is having one valence electron.

11 What is a *valence electron*?

It takes energy to remove an electron from an atom to form a positive ion. This energy is called **ionisation energy**.

12 The first ionisation of a metal can be represented as $M(s) \rightarrow M^+ + e^-$, where M represents an atom of a metal. Write this representation in words.

13 The second ionisation of a metal can be represented as $M^+ \rightarrow M^{2+} + e^-$. Write this representation in words.

Group I: alkali metals Moving down Group I from lithium (Li) to caesium (Cs), the first ionisation energies decrease (i.e. atoms lose electrons more easily and hence form positive ions more easily) and so reactivity will increase. When Group I metals lose their outer, single valence electron they lose a whole electron shell because the single electron was all that was found in the shell.

Melting points and hardness decrease as you move down the group. The size of the atoms will increase moving down the group because extra electron shells are being added (see table 1).

Alkali metals tend to form ionic compounds with non-metals. Because of their high reactivity, they are not found uncombined in nature. The last element in the group, francium (Fr) is radioactive. Group I metals react with water to form hydrogen and the metal hydroxide as products; e.g. solid sodium (Na) reacts with liquid water (H_2O) to give hydrogen (H_2) gas and a solution of sodium hydroxide (NaOH).

14 Write a balanced formula equation showing the reaction between sodium and water.

15 Give the symbols (showing charge) for the ions formed by the Group 1 metals.

16 Describe what happens to the size of the atomic radius as you move down Group I.

17 Why is the ionic radius for a Group I metal less than its atomic radius?

Group II: alkaline earth metals These elements include beryllium (Be) to radium (Ra) and are all harder and less reactive than the alkali metals. Group II metals exhibit the 2+ oxidation state; i.e. they form ions with a 2+ charge.

18 Give the symbols (showing charge) for the ions formed by the Group II metals.

The Group II metals react with water (but not as strongly as the Group I metals) to form hydrogen and the metal hydroxide; e.g. solid calcium (Ca) reacts with liquid water (H_2O) to give hydrogen (H_2) gas and a solution of calcium hydroxide ($Ca(OH)_2$).

First and second ionisation energies decrease moving down the group from beryllium to radium. As a result, reactivity increases moving down the group. Melting points decrease moving down the group.

19 Give two similarities between Group I and Group II metals.

20 Give two differences between Group I and Group II metals.

Group III Boron (B) is classed as a metalloid, aluminium (Al) is classed as a metal and the other Group III elements are metals. Aluminium does not appear to be reactive because it has a protective oxide coating. Aluminium, gallium (Ga), indium (In) and thallium (Tl) all form 3+ ions.

21 As you move down Group III, what do you expect to happen to the size of the atoms of the individual elements? Explain your answer.

22 Would you expect boron to form ionic or covalent bonds? Explain your answer.

Element	Atomic radius ($\times 10^{-9}$ m)	Ionic (M^+) radius ($\times 10^{-9}$ m)
Lithium	0.133	0.060
Sodium	0.157	0.095
Potassium	0.203	0.133
Rubidium	0.216	0.148
Caesium	0.235	0.169

Table 1

Group IV This group consists of the elements carbon (C) to lead (Pb). Carbon is a non-metal that forms covalent bonds. Carbon can exist as the allotropes diamond and graphite, which have different properties. Both allotropes contain only the element carbon, but their lattices are arranged differently. In the diamond lattice, all four valence electrons are involved in strong covalent bonds with neighbouring carbon atoms.

Figure 2 shows the layer structure of graphite. Each individual graphite layer is very strongly bonded. Between each layer there are delocalised electrons. Delocalised electrons are not attached to a specific nucleus and they allow graphite to conduct electricity. Graphite is black and much softer than diamond. It can be used as a lubricant as the individual layers slip over each other easily.

Silicon (Si) is a metalloid that forms mainly covalent bonds. Germanium is also a metalloid and forms covalent and ionic (Ge^{4+}) bonds. Tin is a metal and forms ionic (Sn^{2+} and Sn^{4+}) bonds. Lead is also a metal and forms ionic (Pb^{2+} and Pb^{4+}) bonds.

23 As you move down Group IV, what happens to the character of the elements?

24 Describe the structure of graphite.

25 What feature of graphite allows it to conduct electricity?

26 Copy and complete the sentence: *When an element can exist in different forms in the same physical state, each form is called an _____.*

27 Allotropes of the same substance can be very different because of the way their atoms (and electrons) are arranged. Compare the two carbon allotropes, diamond and graphite.

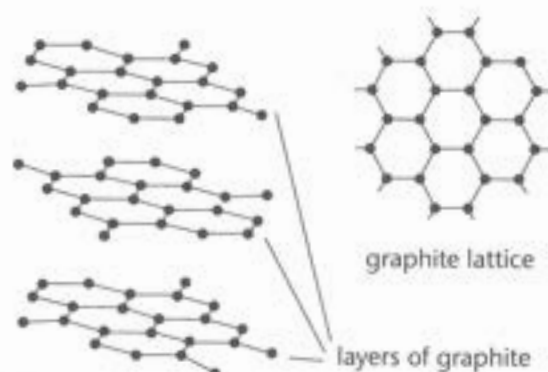


Figure 2

Group V This group has the elements nitrogen (N) to bismuth (Bi). Nitrogen and phosphorus (P), are both definitely non-metals, exhibiting covalent bonding. Nitrogen is a colourless and odourless gas that makes up around four-fifths of the atmosphere. Arsenic (As) and antimony (Sb) are (poisonous) metalloids. Bismuth is a metal. Antimony is a constituent of alloys and semiconductors.

28 Nitrogen exists as diatomic molecules. Explain what is meant by a *diatomic molecule*.

29 Give the formula for a nitrogen molecule.

30 Name the gas that makes up approximately one-fifth of the atmosphere.

31 Explain what is meant by the term *alloy*.

32 Explain what is meant by the term *semiconductor*.

Group VI Oxygen (O) and sulfur (S) are non-metals. The main use of sulfur is making sulfuric acid. Selenium (Se) and tellurium (Te) are metalloids. Selenium is used in the manufacture of solar cells. Polonium (Po), a metal, was the first radioactive element discovered by Marie Curie. A valency of 2 is common in polonium, but others occur.

33 Give the formula for sulfuric acid.

34 Sulfur burns to form the gas sulfur dioxide. Write a balanced formula equation for this reaction.

Group VII The elements are also called halogens ('salt producers'). Their atoms are strong electron acceptors (i.e. strong oxidising agents). Fluorine (F) and chlorine (Cl) are diatomic gases, bromine (Br) is a liquid and iodine (I) is a solid. Their high chemical reactivity makes the halogens corrosive and poisonous. The radioactive element astatine (At) can be made in a nuclear reactor. It resembles iodine but has characteristics that are more metallic. Halogens can form both covalent bonds (with non-metals) and ionic bonds (with metals). Valency is frequently 1 but others can exist.

35 Fluorine reacts with sodium to form sodium fluoride (NaF), a compound used to reduce tooth decay. Write a balanced formula equation for this reaction.

36 Why aren't the halogens found as elements in nature?

37 Seawater contains about 70 parts per million (ppm) by mass of bromine. How much bromine is present in a kilogram of seawater? Show your working.

38 Fluorine is the most reactive halogen. True or false?

Group VIII This group is made up of the elements helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Ra). This group is also called the **noble gases**. The atoms of noble gases are stable and other elements tend to react in a way that produces electron configurations (isoelectronic) similar to those of the noble gases. The noble gases are monatomic; i.e. exist as single atoms. The noble gases either do not react (He, Ne, Ar) or react with great difficulty (Kr, Xe, Rn). The compounds formed are unstable. Radon is radioactive.

39 Explain why the noble gases are so unreactive.

40 The noble gases have also been called the *Group 0 elements*. Give a reason for this.

Transition elements These elements are usually quoted as scandium (Sc) to zinc (Zn), yttrium (Y) to cadmium (Cd) and hafnium (Hf) to mercury (Hg). These elements have incomplete inner electron shells and variable valencies. The transition elements have typical metallic properties.

41 Outline three properties of a typical metal.

Many transition elements (and their compounds) act as catalysts; i.e. they speed up reactions without being used up themselves. Transition metal compounds are often coloured. They have strong metallic bonding and are widely used materials because of their mechanical properties.

42 Complete the table with the names and symbols of these transition metals.

Name	Symbol	Name	Symbol	Name	Symbol
Titanium		Iron		Zinc	
	V		Co		Ag
Chromium		Nickel		Gold	
	Mn		Cu		Pt

Lanthanides In the sixth period there is a set of elements from lanthanum (La) to lutetium (Lu). They are also known as the 'rare earth' metals although they are not particularly rare. They are set apart from the main Periodic Table and have very similar chemical and physical properties.

Actinides In the seventh period there is a similar group of elements beginning with actinium (Ac) and ending in lawrencium (Lr). All of the actinide elements are radioactive. Thorium (Th) and uranium (U) occur in nature. The other actinides are manufactured elements (transuranics), produced only as a result of nuclear reactions.

Periodic properties

For the substances listed in the Periodic Table, the properties usually studied include size, melting point, boiling point and first ionisation energy. If these are plotted against atomic number, trends are revealed as you move from left to right across a period.

Size The diameter of atoms increases as you read down a group. This is due to the increasing number of electron shells. As you move across a period, the diameter decreases. This is because there is a greater attraction between the increasing number of protons in the nucleus and the electrons in the same shell.

Structure The changes in melting point that take place across a period are related to structure. If you study a table of melting points, you will see that there are discontinuities (sudden changes); e.g. between carbon and nitrogen and between silicon and phosphorus. This indicates the change in the structure of the elements from giant lattices to discrete molecules.

43 Explain what is meant by a *discrete molecule*.

44 Mark each of the following statements as true or false.

a The lanthanides are very rare elements.

b The actinides are radioactive elements.

c As you move across a period, the size of the atoms increases.

d The lanthanides resemble each other in their properties.

e Elements are grouped on the Periodic Table according to similarities in their properties.

Over the centuries there have been many theories about **matter**. Matter can be defined as anything that takes up space and has mass or substance. The idea that matter is made up of tiny pieces (particles) is credited to Democritus, a Greek philosopher who lived around 2500 years ago. The technology to test the theory of particles did not exist for many centuries and little thought was given to this concept until John Dalton (1776–1844) revisited this proposal. Dalton used a Greek word, *atomos*, to describe his impression of the smallest particles that could exist. Dalton's concepts were not welcomed by the established scientists of that time.

Dalton's atomic theory can be summarised as:

- All matter is made of atoms
- Atoms cannot be created, nor destroyed nor divided
- All the atoms of a specific element are identical
- The atoms of a particular element are different from the atoms of all other elements (In other words, the atoms of each element are unique.)
- Atoms can combine with other atoms of the same, or different types, to form molecules
- All of the molecules of a specific compound are identical
- Atoms combine in simple, whole number ratios.

- 1 Give a reason why nineteenth century scientists might have been sceptical about the existence of atoms.

We now know that not all of the statements in Dalton's atomic theory are completely correct. In nuclear reactions, matter can be changed to energy. Not all atoms of a given element are identical. Some elements exist in the form of *isotopes* where some atoms of that element have different numbers of neutrons in the nucleus.

- 2 Choose two of Dalton's statements that have been modified since they were originally presented. Explain how modern theories differ from them.
- 3 Which of the following statements fit into Dalton's atomic theory?
 - a All carbon atoms have the same mass.
 - b Sulfur atoms have different properties to carbon atoms.
 - c Chemical reactions involve the rearrangement of atoms.
 - d The mass of reactants in a reaction will be the same as the mass of the products.

Dalton thought of the atom as a small, solid sphere. This model continued to be popular until JJ Thomson (1856–1940) discovered the electron in 1897. Thomson proposed that an atom is a positive sphere of electrification within which there are a regular array of negative electrons of equal value (or equal in number to the positive charges) which gives an overall neutral structure.

- 4 Draw and label a diagram of Thomson's model of the atom.

In 1913 Thomson discovered that some elements were composed of atoms of different mass but with identical chemical properties. These are called *isotopes* (from the Greek word meaning 'place').

In 1911 Ernest Rutherford (1871–1931) devised his own model of the atom after a series of experiments with radioactive sources. The Rutherford (nuclear) atom has the following features:

- Each atom has a centre – a nucleus
- The nucleus is tiny compared to the overall volume of the atom
- The nucleus has most of the mass of the atom
- The nucleus is positively charged
- Negatively-charged electrons rotate around the nucleus
- The charges on the electrons and the nucleus balance to give an electrically-neutral atom.

- 5 Draw and label a diagram of Rutherford's model of the atom.

- 6 Define *isotope*.

Research 7 Some sources call Ernest Rutherford 'The father of nuclear energy'. Explain why.

What do you think? 8 Dalton was a keen player of lawn bowls and this has been linked to his atomic theory. Discuss how these two things could be related.

Rutherford's famous experiment in 1911 helped him devise his model of the atom. He used a radioactive source to fire alpha (α -) particles at a very thin sheet of gold foil. Figure 1 is a simplified diagram of his experiment. When α -particles hit the zinc sulfide screen they cause fluorescence (i.e. light is given off). Rutherford moved the screen to find where the α -particles travelled after they hit the gold foil. He found that almost all of the α -particles passed straight through the foil (which was 6×10^{-7} m thick) but a few were deflected and about one in 20 000 bounced back.

9 This experiment was carried out in a vacuum. Explain what a vacuum is and why the experiment was performed in one.

10 Review Rutherford's model of the atom on the previous page. Explain how the results of the α -particle scattering experiment led him to this atomic model.

11 Suggest why the alpha source was placed in a lead block.

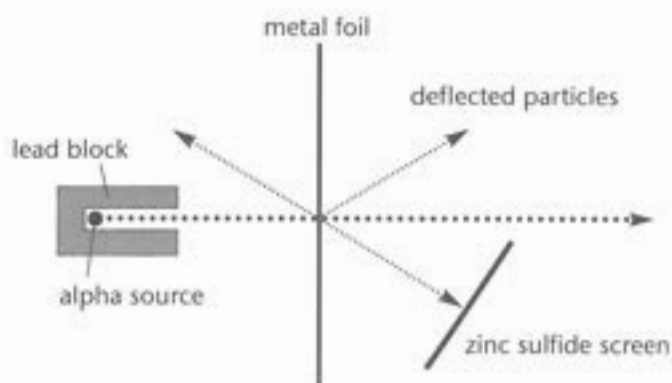


Figure 1

Later experiments showed that the nucleus of an atom is around 100 000 times smaller than its diameter. Rutherford's basic idea of the nuclear atom is still valid, but our understanding of the atom has evolved.

For many years it had been noticed that when elements are heated they emit light energy of specific wavelengths. These atomic spectra were analysed by many scientists. Max Planck (1885–1947) introduced the idea that energy is released (or absorbed) in bursts rather than as a continuous flow. Each of the bursts is a multiple of a very small unit or bundle of energy called a **quantum**. Neils Böhr (1885–1962) won the Nobel Prize for Physics in 1922 for his work on atomic theory. Böhr applied Planck's quantum theory to the orbits of electrons. His discoveries can be summarised as follows:

- The atom has a positively-charged nucleus
- Electrons circle the nucleus
- Electrons can only move on defined paths called stationary states
- These paths (orbits) are separated by at least one quantum
- When electrons move between stationary states, specific energy changes are associated with these movements.

Figure 2 shows one representation of the Böhr atomic model

12 Outline the similarities between the atomic models put forward by Rutherford and Böhr.

13 Outline the differences between the atomic models put forward by Rutherford and Böhr.

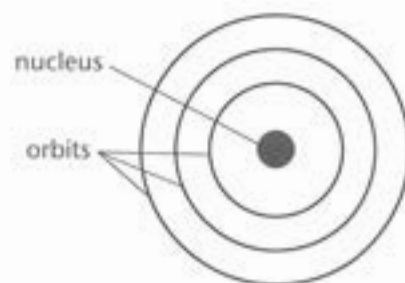


Figure 2

Böhr's model allows for a maximum of 2 electrons in the first orbit and a maximum of 8 electrons in the outer orbit. It works very well in predicting the valency of elements and the formulae of compounds. More sophisticated models have since been developed. These models concentrate on the probability of finding an electron in a particular space.

Figure 3 shows two ways of representing the probability of finding an electron at a specific distance from the nucleus.

14 Describe how the diagrams in figure 3 represent the position of an electron.

15 Discuss the differences between Böhr's model and figure 3.

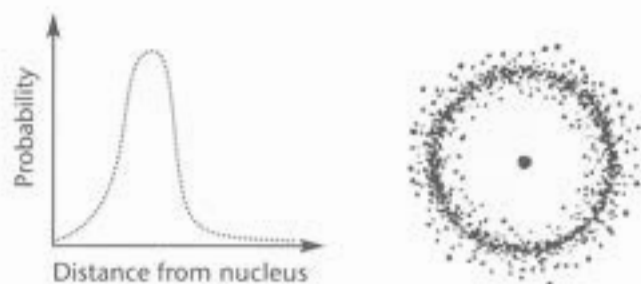


Figure 3

Atomic theory

An atom is the smallest particle of an element that can exist and still behave like the element. Single atoms are far too small to see, even with the most powerful light microscope in existence. An atom is made up of three other smaller particles known as **subatomic particles**. The three types of subatomic particles are protons, neutrons and electrons. Figure 1 shows one model of the atom with these particles.

In the centre of the atom is the nucleus. The nucleus contains protons and neutrons. Protons have a positive (+) electrical charge. Neutrons have no electrical charge; they are neutral. Electrons orbit around the outside of the nucleus. Electrons have a negative (-) electrical charge. Protons and neutrons each have a mass of 1 amu (atomic mass units) and electrons have a mass of approximately 0.0005 amu. One amu is defined as one-twelfth of the mass of one atom of carbon-12. It is a very small unit of mass: 1 amu is equal to 1.66×10^{-24} grams.

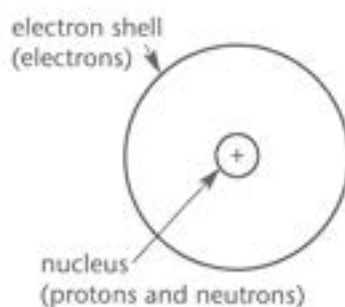


Figure 1

- 1 Define *subatomic particles*.
- 2 Draw up a table to show the above information about the make up of an atom.

In an atom, the number of electrons is equal to the number of protons. This means that atoms are electrically neutral because the positive charges are 'balanced' by the negative charges. Each element has an **atomic number**. This gives the number of protons (which is also the number of electrons). The atomic numbers of each element are shown in the Periodic Table. Hydrogen (H) is the simplest element and has an atomic number of 1. This means it has 1 proton and 1 electron.

Elements also have an **atomic mass** (or mass number). This is the sum of the number of protons and neutrons present in one atom. The number of neutrons can be calculated by subtracting the atomic number from the mass number. For example, chlorine ($^{35}_{17}\text{Cl}$) has an atomic number of 17 and a mass number of 35. This means it has 17 protons, 17 electrons and 18 neutrons ($35 - 17$).

- 3 For $^{55}_{25}\text{Mn}$, give the:
 - a mass number
 - b atomic number
 - c number of electrons
 - d number of protons
 - e number of neutrons.

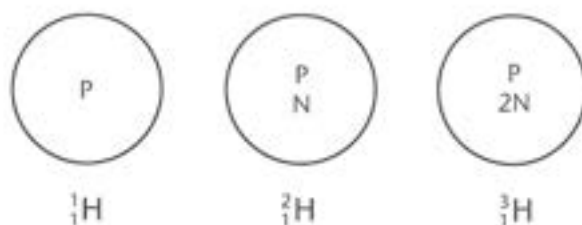


Figure 2

Isotopes

Sometimes atoms of the same element are not identical but have different masses. This is because they have different numbers of neutrons. These atoms are called **isotopes**.

Most hydrogen atoms have 1 proton, 1 electron and no neutrons. Some less common forms of hydrogen have 1 proton, 1 electron and 1 neutron and are called deuterium atoms. Some have 1 proton, 1 electron and 2 neutrons. These are called tritium. Figure 2 shows the isotopes of hydrogen.

The atomic mass of a substance can be calculated if the percentages of isotopes occurring in nature are known. In nature there are two isotopes of chlorine: chlorine-35 ($^{35}_{17}\text{Cl}$) and chlorine-37 ($^{37}_{17}\text{Cl}$). Chlorine-35 has 17 protons, 17 electrons and 18 neutrons. Chlorine-37 has 17 protons, 17 electrons and 20 neutrons. The atomic mass of naturally occurring chlorine is quoted as 35.45 amu. This is because 76% of chlorine atoms are Cl-35 and 24% are Cl-37. Figure 3 shows how the average atomic mass of chlorine is calculated.

- 4 Copper occurs as two isotopes. One isotope has a mass of 63 amu (69%) and the other has a mass of 65 amu (31%). Calculate the average atomic mass of copper. Show your working.

$$\frac{(35 \times 76) + (37 \times 24)}{100} = 35.45$$

Figure 3

- Research 5** Atomic mass units have also been called *daltons*. Find out the origin of this name.

Electronic configuration

In the Bohr model of the atom, electrons are arranged in shells (orbits) around the nucleus. The first shell is the closest to the nucleus. It is the lowest energy level and can hold up to two electrons. The second shell can hold up to a maximum of 8 electrons. When this shell is full, a third shell begins which can also hold up to 8 electrons. When the third shell is full, a fourth one begins and so on until all the electrons fit. The further out a shell is from the nucleus, the higher the energy that each electron in that shell has. An atom in which the outermost (furthest from the nucleus) electron shell has eight electrons (an octet) has a special stability associated with it.

The electronic configuration of an element can be calculated from its atomic number. For example, magnesium (Mg) has an atomic number of 12, which means there are 12 electrons in an atom. It will have 2 electrons in the first shell, 8 electrons in the second shell and 2 in the third shell: $2 + 8 + 2 = 12$.

6 Copy and complete the following table. The first one has been done for you.

	Symbol	Name	No. of electrons	Electronic configuration
	B	Boron	5	2, 3
a	C			
b	F			
c	Ne			
d	Na			
e	S			
f	Ca			

7 What is an octet of electrons?

The least reactive elements are the **noble gases** in Group VIII (0) of the Periodic Table. These elements have atoms with full outer shells (see figure 4). They take part in very few reactions. Atoms with full outer electron shells are more stable than atoms with incomplete energy shells. A full energy shell can be achieved by gaining or losing electrons. This is accomplished by taking, giving up or sharing 'loose' electrons with other atoms where the outer shell does not have the full octet. The electrons in the outermost shell are called **valence electrons**. These electrons are the most easily removed and are the ones that are involved in chemical reactions. Knowing the number of valence electrons can help in predicting how an element will react.

8 Name three noble gases.

9 In which group of the Periodic Table are the noble gases?

10 Why are noble gases unreactive?

11 What are valence electrons?

The atoms of the elements in Groups I to VII have incomplete outer shells. By reacting with each other, atoms can gain, lose or share electrons to form full outer shells (octets) by which they become more stable. Metals tend to lose electrons and non-metals tend to gain electrons. When they do this they form charged particles called ions. Metals form positive ions called cations and non-metals form negative ions called anions.

12 Explain why the loss of an electron from an atom would result in a positively-charged particle.

The noble gases

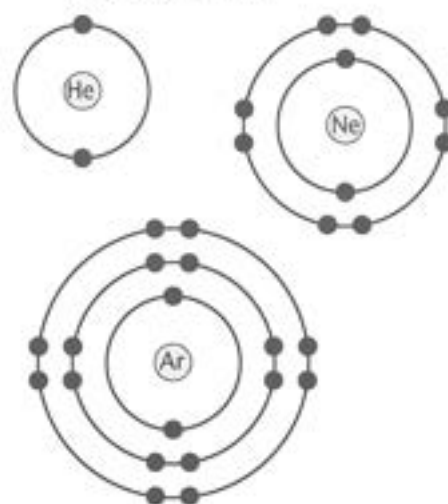


Figure 4

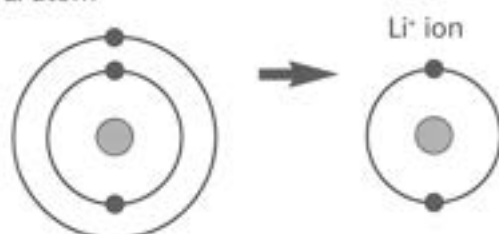
The noble gases (Group VIII) have atoms with full electron shells; because of this, they are unreactive. All other elements have atoms with incomplete outer shells. By gaining, losing or sharing electrons, atoms can achieve full shells and become stable. Metal atoms tend to lose electrons (which have a negative charge), forming positive ions. Non-metals gain electrons, forming negative ions. If an atom loses one electron it forms an ion with a charge of $1+$; if it loses two electrons it forms an ion with a charge of $2+$.

An **ion** is an electrically-charged particle. It has a charge because it contains different numbers of protons (positive) and electrons (negative).

- 1 Give the charge on an ion formed by an atom that has lost three electrons: _____
- 2 Complete the diagrams below to show how each atom would form an ion. Include:
 - the final electronic configuration (how the electrons are arranged in shells)
 - how many electrons have been lost
 - the charge on the ion formed.

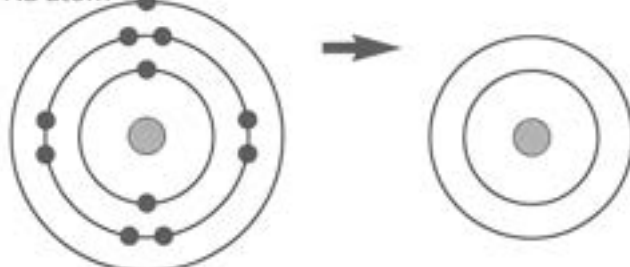
The first one (lithium, Li) has been done for you.

a Li atom

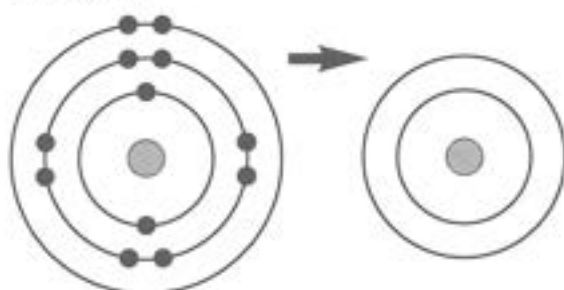


loses one electron

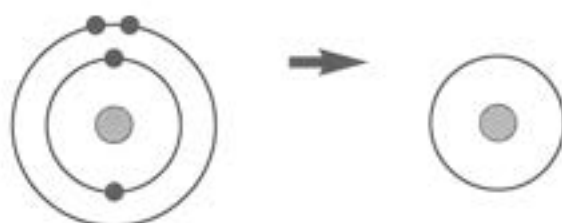
b Na atom



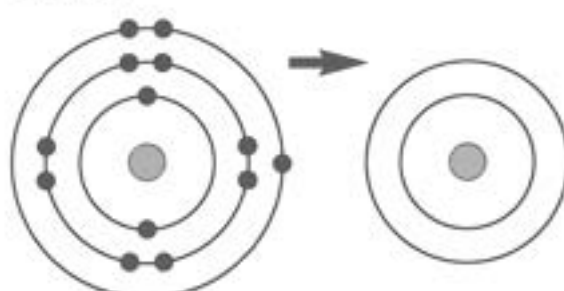
c Mg atom



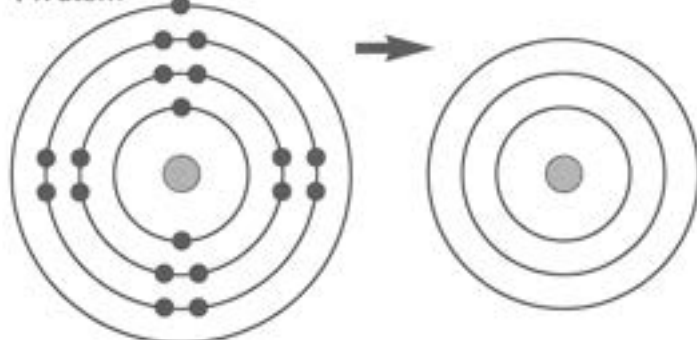
d Be atom



e Al atom



f K atom



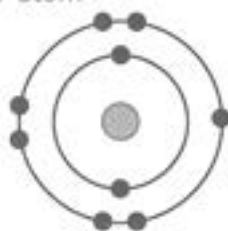
- 3 State two things that the atoms in question 2 have in common.

Non-metals, if they form ions, tend to gain electrons. Their ions have more electrons than protons.

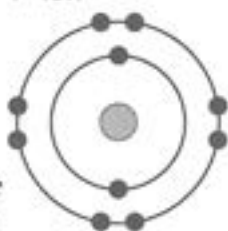
- 4 What is the electrical charge of an ion formed by a non-metal that gains an electron? _____
- 5 Complete the diagrams below to show how each atom would form an ion. Include:
 - how many electrons have been gained
 - the final electronic configuration
 - the charge on the ion.

The first one has been done for you.

a F atom

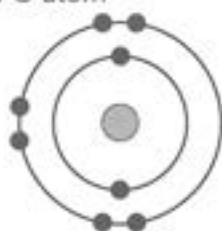


F⁻ ion

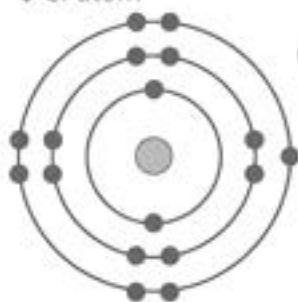


gains one electron

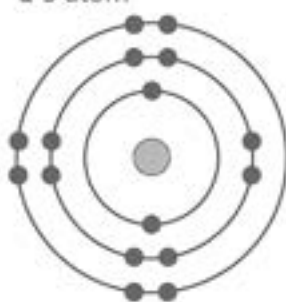
b O atom



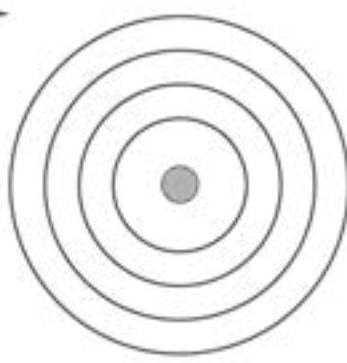
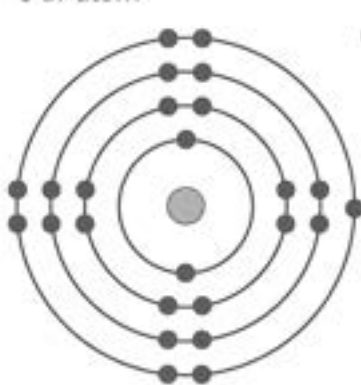
c Cl atom



d S atom



e Br atom



- 6 State two things that the atoms in question 5 have in common.

- 7 Write the name of each element and state the group to which it belongs.

a Li	d Be	g F	j S
b Na	e Al	h O	k Br
c Mg	f K	i Cl	
- 8 What is the relationship between an element's group and the number of electrons it tends to gain or lose?

- 9 Is it correct to state that a metal often forms a bond with a non-metal? Explain your answer.

GN Lewis (1875–1945) was an American chemist and one of the first people to come up with a theory of electronic bonding. He used dots to represent electron structures.

Atoms transfer or share electrons to form bonds. These bonds hold the atoms together. If an atom loses an electron, it forms a positive ion. If it gains an electron it forms a negative ion.

1 What is the electrical charge on an electron?

Metals tend to lose electrons and non-metals tend to gain or share electrons in bonds. The transfer of electrons results in **ionic bonds**. Sharing electrons results in **covalent bonds**. There is a shorthand way of showing how electrons are involved in bonding: ‘**electron dot**’ formulae. These show only the valence (outer) electrons, leaving out the inner shells of electrons. Therefore, there may be many more electrons present but the ‘inner’ levels are not illustrated.

2 When studying bonding, why can the inner shells of electrons be left out?

Figure 1 uses Lewis electron dot structures to show how sodium (Na, atomic number 11, electronic configuration: 2, 8, 1) and chlorine (Cl, atomic number 17, electronic configuration: 2, 8, 7) atoms react to form the ionic compound, sodium chloride (NaCl).

3 What is meant by *electronic configuration*?

4 Copy figure 1 and add labels to describe what occurs.

5 Give the symbol and electronic configuration of potassium (atomic number 19).

6 Give the symbol and electronic configuration of fluorine (atomic number 9).

7 Use Lewis electron dot structures to show how potassium and fluorine react to form potassium fluoride (KF).

8 What type of bonding is present in both sodium chloride and potassium fluoride?

9 Use Lewis electron dot structures to show how magnesium (atomic number 12) and oxygen (atomic number 8) react to form magnesium oxide (MgO).

Electron-dot formulae can also be used to show covalent bonds. Figure 2 shows how chlorine can form chlorine (Cl_2) molecules. Figure 3 shows how oxygen atoms can form an oxygen molecule (O_2).

10 Copy figure 2 and add labels to explain what happens between the atoms.

11 Use Lewis electron dot structures to show how fluorine atoms react to form fluorine (F_2) molecules.

12 Copy figure 3 and add labels to explain what happens between the atoms.

13 Outline the ways electrons are involved in ionic bonding and compare this to the ways electrons are involved in covalent bonding.

Tetrachloromethane (CCl_4) is a covalent molecule. Carbon has an atomic number of 6. Figure 4 shows part of the structure of the tetrachloromethane molecule.

14 Copy figure 4 and show how the electrons are arranged in the molecule.

Research 15 GN Lewis also developed a theory of acids and bases. Outline this theory.

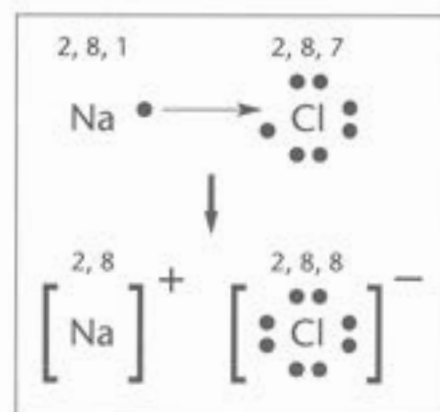


Figure 1

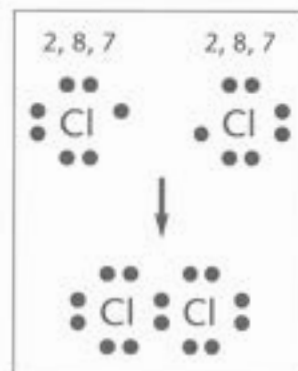


Figure 2

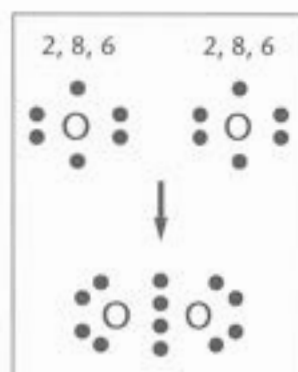


Figure 3

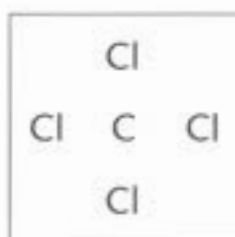


Figure 4

Only the noble gases exist as discrete (separate) atoms. Imagine that a complete outer shell of electrons forms a kind of 'case' around the atom. Then think of that case as complete and enclosing the whole unit. The atoms of all other substances that are not noble gases become bonded in some way as a result of having incomplete outer 'shells'. These incomplete shells need to gain, lose or share a specific number of electrons to form a full outer shell. When the shell is full, the substance reaches the stable electronic configuration of the noble gases. The type of bonding present in a compound will affect its properties (the way it looks, feels and behaves). The main types of bonding are ionic, covalent and metallic.

Ionic compounds

These compounds are also known as **electrovalent compounds**. Ionic compounds are composed of ions (charged particles) and electrically-neutral molecules. Ions are formed by electron transfer between metals and non-metals. The ions that are formed are attracted by the opposite electric charges. These ions quickly arrange themselves into giant structures in which each positive ion is surrounded by negative ions and vice versa. Once ions have been formed, they do not have the same properties as their parent atoms. This is because they now have a different electronic structure to the parent atom and therefore different chemical properties. Ionic compounds are generally solids (often crystalline).

- 1 Circle the ionic compounds in the list below.

NaF, H₂O, MgCl₂, H₂, CaI₂, PCl₃, LiBr, NH₃

In ionic compounds, the positive ions (**cations**) have been formed from metals and the negative ions (**anions**) from non-metals. For example, sodium chloride, NaCl, is made up of sodium (Na⁺) and chloride (Cl⁻) ions. Its structure is a repeating three-dimensional lattice that is held together very tightly because the positively-charged sodium ions are attracted to the negatively-charged chloride ions.

Each ion is surrounded (in three dimensions) by six oppositely-charged ions. The shape of the lattice is a cube. The attractive force between ions is large (and non-directional) and a lot of energy is needed to move the ions apart. This tightly-packed structure makes the bonds harder to break, resulting in high melting and boiling points. Figure 1 shows part of one layer of an ionic lattice.

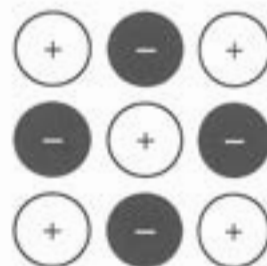


Figure 1

- 2 Redraw figure 1 to show how sodium and chloride ions might be arranged in a layer of an ionic lattice.
- 3 Add to your diagram to show how the next layer of ions might be arranged.

The formula for an ionic compound is an **empirical formula**; it shows the ratio of ions present, not the actual numbers.

- 4 Why is the empirical formula given for an ionic compound and not the actual numbers of ions present?
- 5 Give the electrical charge on an anion.
- 6 Give the electrical charge on a cation.

When looking at the solubility of substances, the phrase 'like dissolves like' is helpful. Many ionic compounds are soluble in water, which is a polar solvent. Polar substances have electrically-charged ends. Figure 2 shows a simple polar substance. One end is slightly positive (shown as δ^+) and the other end is slightly negative (shown as δ^-). The symbol δ (delta) stands for a small change. Ionic compounds are usually insoluble in organic (carbon-based) solvents, because these solvents are covalent substances, not polar substances.

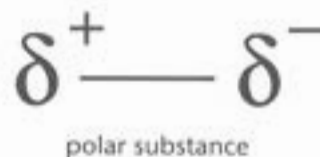


Figure 2

An electric current is a flow of electrons or ions. The outer electrons in ionic compounds are used to give the ions their noble gas type structures. These electrons cannot move away from their ions. Although ionic compounds are composed of charged particles, they do not conduct electricity when they are solid. When molten (fused) or in solution, the ions are no longer held together as strongly and they are free to move. If electrodes are placed in such a liquid, the ions allow a current to flow and this is accompanied by decomposition (breakdown) of the substance. This is called **electrolysis**.

- 7 What does it mean to say an ion in a compound has reached a noble gas type structure?
- 8 Name two types of bonding apart from ionic.
- 9 Name the only types of elements that exist as separate (non-bonded) atoms.
- 10 Why do ionic compounds tend to have high melting points?

Figure 3 shows how the ions move apart when an ionic compound is dissolved. (Note that the water molecules are not shown in the diagram.) If positive and negative electrodes are placed in an ionic solution, the ions will be affected.

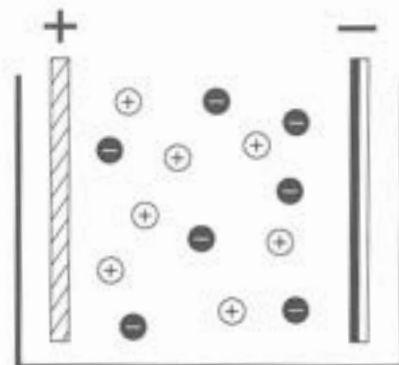


Figure 3

- 11 Using figure 3 as a model, draw a diagram to show how ions would move in a solution if electrodes were present.
- 12 Why can ionic compounds only conduct electricity when they are in liquid form?
- 13 Match the following terms with the definitions below:
cation, electrolysis, empirical, ionic, octet, polar, solvent, valence
 - a The type of bond formed from the electrostatic attraction between oppositely-charged ions: _____
 - b An ion that would be attracted to the cathode in electrolysis: _____
 - c Outermost electrons that are transferred or shared in bonding: _____
 - d The process that uses electricity to split up an ionic compound when it is in liquid form: _____
 - e A formula that shows the ratio of elements present: _____
 - f A liquid that can dissolve other substances: _____
 - g The full outer shell of electrons that is present in the noble gases from neon to radon: _____
 - h A substance that is not ionic but has ends that have a slight difference in electrical charge: _____

Covalent substances

Compounds that contain only non-metals usually consist of molecules. They can be molecules of single elements (e.g. oxygen, chlorine, carbon) or compounds of different elements. Covalent substances can be divided into two groups: small (discrete) molecules and giant molecules (macromolecules/covalent lattices). In both cases, the atoms within a molecule are bonded to each other by strong covalent bonds. Covalent bonds contain shared electrons. Small molecular substances often have between two and twenty atoms per molecule. Giant molecules have many millions of atoms per molecule. Remember, in ionic compounds, one atom tends to give up 'spare' electrons to another atom that only needs a few electrons to fill its outer shell. In covalent bonding, one atom does not 'take' electrons from another, they are shared.

Small molecules are discrete; they tend to be separate and can move independently of each other. They have low fixed points (melting points and boiling points). This is because there are only relatively small attractive forces between electrically neutral molecules. Tetrachloromethane is a typical molecular covalent substance. It is a liquid with a low boiling point. The structure of tetrachloromethane is shown in figure 4.

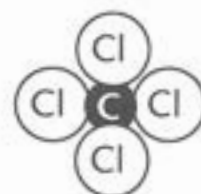


Figure 4

- 14 What is the formula of tetrachloromethane?
- 15 Review your notes on the previous page explaining how ionic compounds are formed, then look at the tetrachloromethane molecule in figure 4.
 - a Are there any metals present in this molecule?
 - b What does the composition of the tetrachloromethane molecule suggest to you about the type of bonding occurring between the atoms (ionic or covalent)? Explain your answer.

Substances made up of molecular covalent molecules often have a smell; ionic and macromolecular substances generally do not. Visualise a smell as molecules that get up your nose; from this it is clear that molecular covalent substances can put molecules into the vapour stage more easily than other substances can. (It is breathing in vapours that allows us to smell things.)

16 Why do molecular covalent substances have low boiling points?

Diamond is made up of carbon atoms arranged in a lattice. Figure 5 shows the structure of diamond. Each carbon atom is attached to another four carbon atoms. The number of the bonds and the strength of the bonds make diamond a very hard substance with a high melting point.

17 Discuss the statement: *A perfect diamond is one single, giant molecule.*

It is much easier to melt a substance that has a small molecular structure than a substance that has a covalent lattice. There is no need to break the strong covalent bonds inside the molecules; rather, separate molecules that are only weakly attracted to each other. It is important to distinguish between **intermolecular forces** and **intramolecular forces**.

Intramolecular forces are those inside the molecule and they are strong covalent bonds (*intra* means within, inside). Intermolecular bonds are between individual molecules and they are weak (*inter* means between). Figure 6 shows the bonding in methane (natural gas), a molecular covalent substance.

18 Give the formula for methane.

19 Melting and boiling points are sometimes called 'fixed points', because they occur at fixed temperatures. Will methane have low or high fixed points? Explain your answer.

Covalent compounds do not conduct electricity in solid, liquid or gaseous states. There are no ions or free electrons to carry a current. Many covalent substances are insoluble in water (a polar substance), but soluble in other covalent liquids such as ethanol (C_2H_5OH) and tetrachloromethane (CCl_4).

20 What type of bonding is present in the ingredients of solid air-fresheners? Explain your answer.

21 Give the names and formulae of two covalent liquids.

22 Explain why covalent substances do not conduct electricity.

23 Suggest why many covalent substances are soluble in other covalent liquids.

24 In chemistry, what does it mean when a compound is said to be 'organic'?

Figure 7 shows two ammonia molecules. Ammonia is a covalent compound. It is a gas at low temperatures but is easily liquefied.

25 Give the formula for ammonia.

26 Copy figure 7. On your diagram show the intermolecular and intramolecular bonding. (Hint: see figure 6.)

27 Ammonia is very soluble in water. Does this mean that ammonia is polar or non-polar? Explain your answer.

28 Make a list of the similarities and differences between ionic and covalent bonding.



Figure 5

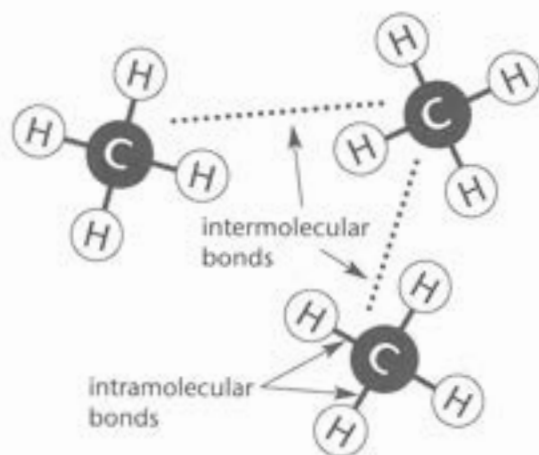


Figure 6

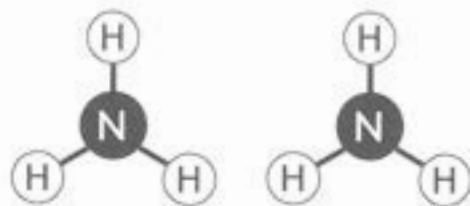


Figure 7

Metals can be recognised by their characteristic physical properties.

- 1 Name three typical properties of metals.

Metals form two types of bonds: ionic bonds and metallic bonds. They form ionic bonds with non-metals. In ionic bonding, metals obtain noble gas electron structure by transferring electrons to non-metals.

- 2 What type of ions are formed by metals in ionic bonds?

In metals, the bonding between atoms is often strong. Metal structures contain billions of atoms packed together very tightly. The atoms are packed together in simple structures. These patterns can be modelled by dropping many marbles into a small box. Many metals are dense; they have atoms that are heavy and pack closely together. When metal atoms bond together, they do not form a noble gas type structure.

- 3 Explain what is meant by *noble gas type structure*.

Electrons are still shared in metallic bonds, but these bonds are different from covalent bonds in one important way. In a covalent bond, the electrons spend their time between the two atoms that are bonded. When metal atoms share their electrons, these electrons are free to move anywhere within the metal structure. There is a 'sea of electrons' surrounding the positive metal cations (see figure 1). Another way to visualise this is to imagine metals as three-dimensional lattices of positive ions in a 'sea' of valence electrons. The delocalised outer-shell electrons in the lattice are extremely mobile and can readily move through the lattice when excited by either thermal or electrical energy. Metallic bonding is strong; hence metals are often hard and have high melting points. However, layers in the metal lattice can be forced to 'slip over' each other; this accounts for the malleability and ductility of metals. Mixtures of different metals are called alloys.

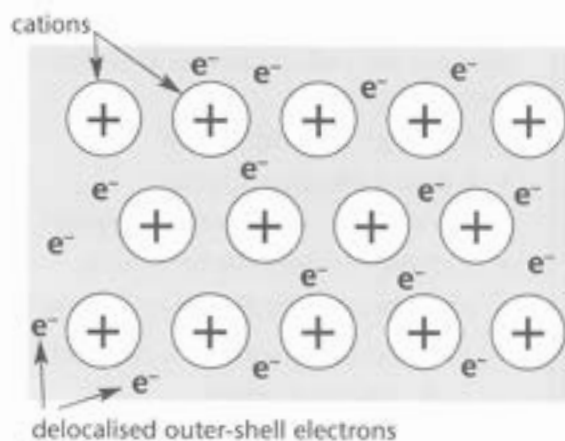


Figure 1

- 4 What are *valence electrons*?
- 5 Give another name for *thermal energy*.
- 6 Define *ductile*.
- 7 Explain why metals are good electrical conductors.
- 8 Use figure 2 to describe how the electrical conductivity of a metal varies with temperature.

- 9 Match the following terms with the definitions below:
alloy, cation, lattice, malleable, metallic bonding

- a The electrostatic force of attraction that neighbouring cations have for the delocalised electrons surrounding them: _____
- b A substance made up of two or more metals or a metal and a non-metal mixed together, usually when molten: _____
- c An atom or group of atoms that carries a positive electric charge: _____
- d Can have its shape changed by beating with a hammer or by the pressure of rollers: _____
- e The regular geometrical arrangement of particles in a metal or crystal: _____

- 10 Metals can form two types of bonds: metallic and ionic. Explain the differences between these types of bonds.

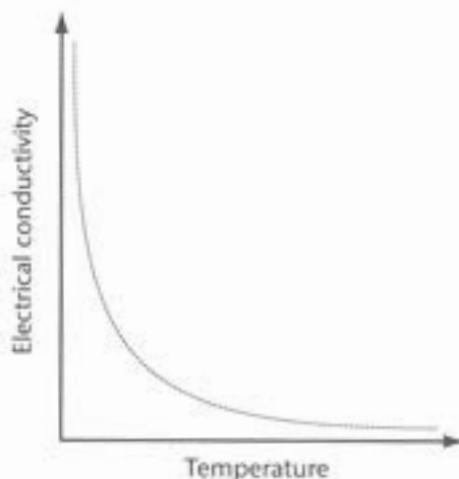


Figure 2

Chemistry is a science that studies the composition and properties of substances and how those substances interact and transform to create new substances. **Physical changes** involve the changing of a substance from one state to another without altering the chemical nature of the substance. For example, melting ice and boiling water are physical changes. The bonds between the water molecules are affected but the molecules remain unchanged. In a **physical change**, no new substance is formed and it is relatively easy to reverse the change.

1 What physical change is the 'opposite' of boiling? _____

2 Why is electricity passing through a light globe a physical change?

3 Give an example of a physical change not already mentioned. _____

Chemical reactions occur when atoms are rearranged to form new substances (compounds). In a chemical change, new substances are formed when atoms form new bonds with each other. Chemical changes are called **chemical reactions**. Chemical reactions happen when one or more of the following take place:

- A change in temperature
- A new substance appears
- The original substance disappears
- A change is difficult to reverse.

4 State whether the following are physical changes or chemical changes.

a Soup from a can is warmed in a saucepan. _____

b Ice-cream melts. _____

c A tomato ripens. _____

d Sugar dissolves in a cup of coffee. _____

e Milk turns sour. _____

f Water is added to a glass of cordial. _____

g A mixture of flour, water and yeast is heated to make bread. _____

5 Name another chemical reaction not already mentioned. _____

Energy is often needed to start a chemical reaction. A match must be struck against the side of a matchbox to ignite it. A naked flame must be placed next to gas coming from a Bunsen burner to light it. These are examples of **activation energy**. Not all chemical reactions occur at the same rate (speed). Magnesium (Mg) and gold (Au) are both shiny metals. A piece of magnesium can burn in a few seconds but gold does not burn – it may take centuries for a gold coin to even tarnish (go dull).

6 Explain why cooking an egg is classified as a chemical reaction.

7 What type of energy is needed to charge a car battery? _____

8 What types of energy are given out by a lit Bunsen burner? _____

9 Give a new example of a very fast chemical reaction. _____

10 Give a new example of a very slow chemical reaction. _____

11 What is *activation energy*? _____

12 When a candle burns, both physical changes and chemical reactions occur.

a Describe a chemical change when a candle burns. _____

b Describe a physical change when a candle burns. _____

13 Explain why chemical changes are harder to reverse than physical changes.

In a chemical reaction, the substances are divided into two groups: **reactants** and **products**. The reactants are the materials you begin with and they react with each other. The products are the new substances produced during the reaction. Reactions are not all the same; the reactants and products can be a mixture of solids, liquids and gases. Some reactions give out heat (are **exothermic**). Other reactions take in heat from their surroundings (are **endothermic**).

14 In a reaction, what is meant by:

- a reactants? _____
b products? _____

As a reaction proceeds, the concentration of reactants decreases and the concentration of products increases. The rate of reaction can be calculated by measuring the change in either reactants or products against time. This can be expressed graphically, as shown in figures 1 and 2. The x-axis is time and the y-axis is concentration.

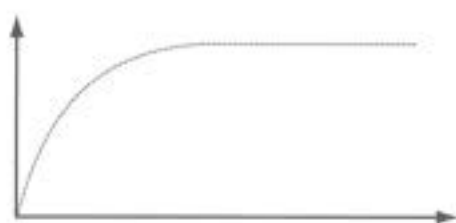


Figure 1



Figure 2

15 Describe what is happening in:

- a Figure 1 (products). _____
b Figure 2 (reactants). _____

16 Explain what is meant by the following terms.

- a Exothermic _____
b Endothermic _____

Three common types of reactions are **synthesis**, **combustion** and **decomposition**. Synthesis involves the formation of a compound from the elements that make it up. Combustion is sometimes described as burning; an element combines rapidly with oxygen, gives out heat and forms an oxide. In decomposition, a compound is broken down into simpler substances.

17 For each reaction, write a word equation and classify it as synthesis, combustion or decomposition.

- a $2 \text{HgO (s)} \rightarrow 2 \text{Hg (l)} + \text{O}_2 \text{(g)}$

- b $\text{H}_2 \text{(g)} + \text{Br}_2 \text{(g)} \rightarrow 2 \text{HBr (g)}$

- c $\text{CH}_4 \text{(g)} + 2\text{O}_2 \text{(g)} \rightarrow \text{CO}_2 \text{(g)} + 2\text{H}_2\text{O (l)}$ (CH₄ = methane)

- d $\text{N}_2 \text{(g)} + 3 \text{H}_2 \text{(g)} \rightarrow 2 \text{NH}_3 \text{(g)}$ (NH₃ = ammonia)

- e $\text{CaCO}_3 \text{(s)} \rightarrow \text{CaO (s)} + \text{CO}_2 \text{(g)}$

18 Give an everyday example of synthesis, combustion and decomposition.

19 In a chemical reaction, the number of atoms in the reactants equals the number of atoms in the products. True or false?

If a compound contains only two elements its name usually ends in '-ide'. In a compound that contains a metal and a non-metal, the metal comes first in the compound's name. For example, iron (III) oxide, Fe_2O_3 , contains the metal iron, Fe, and the non-metal oxygen, O. Sodium chloride (NaCl) contains the metal sodium, Na, and the non-metal chlorine, Cl.

These are some name changes for elements present in compounds:

F, fluorine → fluoride

I, iodine → iodide

Br, bromine → bromide

O, oxygen → oxide

N, nitrogen → nitride

S, sulfur → sulfide

- 1 Name two things that the elements above have in common.
- 2 Copy and complete the table. The first one has been done for you.

Compound	Elements present	Symbols for elements
magnesium oxide	magnesium and oxygen	Mg, O
copper chloride		
zinc oxide		
iron sulfide		
sodium iodide		
magnesium nitride		
potassium bromide		

Often in chemistry atoms form groups that stay together during reactions, acting in a similar way to a single atom. For example, when four oxygen atoms bond with a sulfur atom, a sulfate ion, SO_4^{2-} , is formed. Now consider magnesium sulfate (MgSO_4) in a reaction. This compound can 'break up' to form a magnesium ion (Mg^{2+}) and a sulfate 'unit' with the five atoms that form the sulfate unit remaining intact. A group of atoms that stays together in chemical reactions is called a **radical**. If more than two elements are present in a compound and one of these is oxygen, then the compound's name usually ends in '-ate'. These radicals (groups of atoms) are common parts of compounds:

$-\text{SO}_4$ = sulfate

$-\text{CO}_3$ = carbonate

$-\text{NO}_3$ = nitrate

$-\text{PO}_4$ = phosphate

Naming compounds that contain radicals is not as confusing as it may first appear. The metals keep most (or a recognisable part) of their names and the presence of oxygen in the radical group usually denotes an '-ate' ending to the name.

- 3 Copy and complete the following table. The first one has been done for you.

Formula	Elements present	Name of compound
CaCO_3	calcium, carbon, oxygen	calcium carbonate
CuSO_4		
FePO_4		
AgNO_3		
CuCO_3		
BaSO_4		
NaNO_3		

- 4 What elements are found in the hydroxide, $-\text{OH}$, radical?

Some chemicals have 'common' names. NH_3 is usually called *ammonia* (rather than nitrogen hydride) and H_2O is referred to as *water* (not hydrogen oxide). *Vinegar* is a dilute solution of ethanoic (acetic) acid and has the formula CH_3COOH .

5 Name the elements present in:

- a ammonia _____
- b water _____
- c vinegar _____

Working with formulae

The formula of a compound tells the type and ratios of atoms present. Water has the formula H_2O , which means there are two atoms of hydrogen and one atom of oxygen present in the simplest unit of water. Calcium sulfate has the formula CaSO_4 and so has one calcium atom, one sulfur atom and four oxygen atoms in one calcium sulfate particle.

If there are brackets in the formula with a number subscript outside, then the atoms in the brackets are multiplied by that number. For example, magnesium hydroxide, $\text{Mg}(\text{OH})_2$, has one magnesium atom, *two* oxygen atoms and *two* hydrogen atoms. The brackets show that the hydroxide unit is a radical. Two hydroxide units bond with one magnesium atom to form magnesium hydroxide. Magnesium has a valency of +2 and each hydroxide unit has a valency of -1, so two hydroxides are required to create an electrically-neutral compound.

6 Write the name of each compound and give the names and numbers of atoms present in one unit of that compound.

- a MnO_2 _____
- b CoCl_2 _____
- c MgSO_4 _____
- d AlCl_3 _____
- e $\text{Zn}(\text{OH})_2$ _____
- f CuCO_3 _____
- g Al_2O_3 _____
- h $\text{Ba}(\text{NO}_3)_2$ _____
- i K_3PO_4 _____

Valency

The valency of an element in a covalent compound is the number of bonds formed by one atom of the element in one molecule of the compound. The usual valency of an element is either its group number (given in the Periodic Table) or, if this is greater than 4, 8 minus the group number. The valency of individual ions in ionic compounds is the charge on that ion.

Thus the valency of Group I elements is 1; the valencies of Group II, III and IV elements are 2, 3 and 4 respectively; and that of Groups V, VI and VII are 3 (8 - 5), 2 (8 - 6) and 1 (8 - 7) respectively.

7 Why is no valency given for Group VIII?

In ionic compounds, the metal has a positive charge and the non-metal has a negative charge. (Remember, in ionic compounds there is a transfer of electrons between atoms; in covalent substances electrons are shared.)

The sum of the valencies of the atoms in a compound must be equal. This rule can be used to work out the formulae of compounds.

Consider silicon (Si, Group IV) and oxygen (O, Group VI). We can work out that silicon has a valency of 4 and oxygen has a valency of 2. Therefore, two oxygen atoms are needed to 'balance' one silicon atom and so the formula is SiO_2 .

A different approach is to write the valency below each atom and cross multiply (see figure 1). This gives the formula Si_2O_4 , which we reduce by the lowest common denominator (in this case, divide by 2) to give SiO_2 .

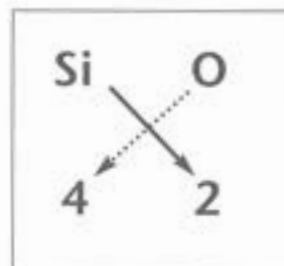


Figure 1

8 Use the procedure shown in figure 1 to work out the formulae of the covalent compounds formed between these non-metals. Include the name of each compound.

- phosphorus and chlorine
- carbon and oxygen
- boron and oxygen
- sulfur and carbon
- nitrogen and hydrogen
- hydrogen and sulfur
- silicon and fluorine

9 Give the general name for groups such as sulphate SO_4^{2-} , carbonate CO_3^{2-} , and nitrate NO_3^- .

10 Mark each statement as true or false.

- A group of atoms that acts as a single unit in a reaction is called a radical.
- When a metal combines with a non-metal the name of the compounds ends in '-ide'.
- When a metal and a non-metal combine the non-metal is always named first in the compound.
- Valency charges are always negative.
- Valency gives an indication of the number or type (single, double, triple) of bond that can be formed in a compound.
- In ionic compounds the metal has a positive charge and the non-metal has a negative charge.

11 State how many atoms of each substance are found in the simplest unit of each compound. The first one has been done for you.

	Compound	Formula	Type and number of atoms
	sodium silicate	Na_2SiO_3	2 sodium, 1 silicon, 3 oxygen
a	calcium hydroxide	Ca(OH)_2	
b	calcium phosphate	$\text{Ca}_3(\text{PO}_4)_2$	
c	zinc nitrate	$\text{Zn(NO}_3)_2$	
d	sodium carbonate	Na_2CO_3	
e	potassium fluoride	KF	
f	copper sulfate	CuSO_4	
g	ammonium sulfate	$(\text{NH}_4)_2\text{SO}_4$	

Formulae of ionic compounds

Ionic compounds are formed between metals and non-metals. Remember, the overall charge on an ionic compound is zero, so the positive charges of metals must be balanced by an equal number of negative charges from non-metals.

12 Give the valency of:

- Group I elements.
- Group II elements.
- Group III elements.

Transition means the change from one stage to another. Transition elements are those found in the columns of the Periodic Table where the transition is made from the metals to non-metals (e.g. scandium to zinc, yttrium to cadmium, hafnium to mercury). Transition elements have variable valencies. The valency of a transition metal in a particular compound is usually shown by Roman numerals.

Copper (II) sulfate means that copper is present as Cu^{2+} in this compound; iron (III) oxide means that iron is present as Fe^{3+} in this compound.

EXAMPLES

- i What is the formula of the compound formed between sodium (Group I metal) and chlorine (Group VII non-metal)?

Sodium forms the Na^+ ion and chlorine forms the Cl^- ion. The ratio must be 1:1, so the formula is NaCl .

- ii What is the formula of the compound formed between magnesium (Group II metal) and oxygen (Group VI non-metal)?

Magnesium forms the Mg^{2+} ion and oxygen forms the O^{2-} ion. The ratio is 2:2; reduce by the lowest common multiple to 1:1. The formula is MgO .

- iii What is the formula of iron (III) sulfate?

Iron (III) is Fe^{3+} . The sulfate radical has a formula of SO_4^{2-} . In order to achieve neutral charge, two Fe^{3+} ions are needed to balance the charge on three SO_4^{2-} ions. (Or follow the cross multiplication method as shown in figure 2.) Hence the formula is $\text{Fe}_2(\text{SO}_4)_3$.

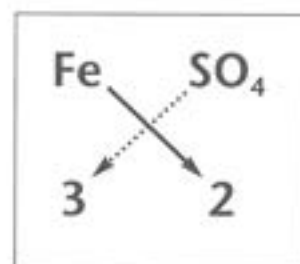


Figure 2

- 13 Copy and complete the following grids by writing the appropriate chemical formulae. The first one has been done for you.

Note that the ammonium radical (NH_4^+) is not a metal but behaves as one when writing formulae.

	Chlorine: chloride, Cl^-	Oxygen: oxide, O^{2-}	Sulfur: sulfide, S^{2-}	Nitrogen: nitride, N^{3-}
Sodium, Na^+	NaCl			
Magnesium, Mg^{2+}				
Iron (III), Fe^{3+}				
Ammonium, NH_4^+				

	Hydroxide: OH^-	Sulfate: SO_4^{2-}	Carbonate: CO_3^{2-}	Phosphate: PO_4^{3-}
Lithium, Li^+				
Calcium, Ca^{2+}				
Aluminium, Al^{3+}				
Ammonium, NH_4^+				

- 14 In the formula iron (II) oxide, what is signified by the Roman numerals?
- 15 'Metals in the middle of the Periodic Table, which have variable valencies.' What does statement define?
- 16 'A number assigned to an element that dictates how many atoms of a given element it can combine with.' What does statement define?

Atoms in chemical compounds have a characteristic combining power or **valency**. This involves the number of electrons in an incomplete outer shell and can be defined in several ways:

- It represents the charge on an ion
- It is equal to the number of electron pairs shared in a molecule
- It is the number of electrons transferred when forming an ion.

1 Use your own textbook to find an alternative definition for *valency*.

For elements in Groups I, II, III, IV and 0, the group valency is the same as the group number (e.g. Group I elements have a valency of 1). For elements in Groups V, VI & VII there are two group valencies. One is the same as the group number and the other more likely valency is 8 minus the group number.

2 The Periodic Table involves groups and periods. Use examples of elements to explain the difference between groups and periods.

3 What is the most likely valency of a Group VI element? Explain your answer.

4 Learn the name, formula and charge on each ion shown in table 1.

+1 valency	+2 valency	+3 valency	+4 valency
lithium, Li^+ sodium, Na^+ potassium, K^+ copper (I), Cu^+ silver, Ag^+ ammonium, NH_4^{+*}	magnesium, Mg^{2+} calcium, Ca^{2+} barium, Ba^{2+} zinc, Zn^{2+} copper (II), Cu^{2+} mercury, Hg^{2+} iron (II), Fe^{2+} nickel (II), Ni^{2+} tin (II), Sn^{2+} lead (II), Pb^{2+}	aluminium, Al^{3+} iron (III), Fe^{3+} chromium (III), Cr^{3+}	tin (IV), Sn^{4+} lead (IV), Pb^{4+}

*The ammonium ion is not a metal ion, but has certain chemical similarities with metal ions.

Table 1

5 Name two things that the ions in table 1 have in common.

6 Give the formula for the ammonium ion.

7 What is the valency of the ammonium ion?

8 Give the two possible formulae for a copper ion.

The **transition metals** (also called the d-block metals) can exhibit more than one valency. These elements are found in the middle of the Periodic Table.

9 Use a textbook to give another definition for *transition metals*.

10 Give two examples of transition metals.

11 Look up the word *transition* in a dictionary. Now explain how you think the term *transition metal* came to be assigned to the transition metal elements.

12 Give your own definition for *valency*.

Radicals

The valency of a particular ion is often shown in Roman numerals inside brackets. Non-metals tend to form negative ions. Groups of non-metal atoms may combine to form **radicals**.

Some examples of radicals are sulphate SO_4^{2-} , carbonate CO_3^{2-} and nitrate NO_3^{2-} . Radicals have also been described as **polyatomic ions**. Table 2 shows some common ions, some of which are radicals.

A **radical** is a charged group of atoms in compounds that always exists as a group.

-1 valency	-2 valency	-3 valency
hydroxide, OH^- nitrite, NO_2^- nitrate, NO_3^- ethanoate, CH_3COO^- hydrogencarbonate, HCO_3^- hydrogensulfate, HSO_4^- hypochlorite, ClO^- permanganate (VII), MnO_4^- cyanide, CN^- hydride, H^- chloride, Cl^- bromide, Br^- fluoride, F^- iodide, I^-	carbonate, CO_3^{2-} sulfate, SO_4^{2-} sulfite, SO_3^{2-} hydrogenphosphate, HPO_4^{2-} dichromate, $\text{Cr}_2\text{O}_7^{2-}$ chromate, CrO_4^{2-} oxide, O^{2-} sulfide, S^{2-} oxalate $\text{C}_2\text{O}_4^{2-}$	phosphate, PO_4^{3-} nitride, N^{3-} phosphide, P^{3-}

Table 2

13 State two things that the ions in table 2 have in common.

14 Write the formula for each ion.

- hydrogen phosphate
- hydrogencarbonate
- sulfate
- chromate

15 Give the names and numbers of atoms present in each ion.

- oxalate
- phosphate
- carbonate

16 Define a *radical*.

17 Give an example of a substance that contains a radical.

18 Give the type of charge that is likely to be found on ions of:

- metals
- non-metals

19 From Table 2, give the names and symbols for two ions that are not radicals according to the definition in the grey box.

Formulae of ionic compounds

The formula of an ionic compound represents the relative numbers of each ion present. Formulae can be worked out by balancing the charges of the ions. For example, in calcium chloride the calcium ion (Ca^{2+}) has a charge of 2+ and the chloride ion (Cl^-) has a charge of 1-. Thus, the simplest unit of calcium chloride requires one calcium ion to balance two chloride ions. Therefore the formula of calcium chloride is CaCl_2 .

20 Work out the formula for each compound.

- | | | |
|------------------------|-------------------------|----------------------|
| a sodium hydroxide | f ammonium carbonate | k lead (IV) sulfide |
| b calcium hydroxide | g iron (III) oxide | l potassium chromate |
| c magnesium oxide | h barium nitrate | m barium phosphide |
| d copper (I) chloride | i copper (II) sulfate | n lithium cyanide |
| e copper (II) chloride | j nickel (II) phosphate | o aluminium oxide |

21 Copy and complete the table. The first one has been done for you.

Common name	Scientific name	Formula
table salt	sodium chloride	NaCl
		NaHCO_3
	calcium hydroxide	Ca(OH)_2
gypsum	calcium sulfate	
epsom salts		MgSO_4
Condy's crystals	potassium permanganate	
Glauber's salt		Na_2SO_4
caustic soda	sodium hydroxide	
saltpetre		KNO_3
potash		K_2CO_3

22 Give the name and formula for the radicals present in each substance.

- epsom salts
- saltpetre

23 Mark each statement true or false.

- The ammonium ion contains metals.
- Valency is the number of atoms from an atom involved in bonding.
- Transition metals can have more than one valency.
- Radicals only contain one type of atom.
- Radicals act much like a single (large) atom in reactions.
- The overall electrical charge in an ionic compound is zero.
- In the name of a compound containing a metal, the name of the metal comes last.
- The common name of a chemical usually tells you something about the formula.

Organic chemistry

Organic chemistry is the study of carbon compounds. At one time, the term 'organic compound' meant a compound that was found in plant or animal materials, such as sugars, fats and proteins. All of these compounds contain carbon. Now the term organic compounds is used for almost all carbon compounds, whether they come from plants and animals or are made in a laboratory. However, compounds such as carbonates (CO_3^{2-}) and carbon dioxide are not usually described as organic compounds.

1 What is the formula for carbon dioxide?

About 5 million chemical compounds are known to occur on Earth. Over 4 million of these are carbon compounds. However, the Earth contains less than 0.01 % by mass of carbon. This huge discrepancy between the proportion of compounds that contain carbon and the proportion of carbon in the Earth indicates that carbon atoms have some very unusual bonding properties. The bonding in most organic substances is covalent.

2 Does covalent bonding involve sharing electrons or the transfer of electrons?

Carbon is in Group IV of the Periodic Table and has four electrons in the outer shell. Because of this, it tends to form *four* covalent bonds. The four electrons available in the outer shell also allows for a much larger number of possible combinations with other atoms. This enables carbon to form many different molecules and explains why it is present in so many compounds.

The bonds form a three-dimensional tetrahedral structure that is often simplified to a two-dimensional view as shown in figure 1, which illustrates the structural formula of a methane molecule. The **structural formula** shows how the atoms are arranged in a molecule (think of a tripod). The **molecular formula** shows the number and type of elements present in each molecule.

3 What is the molecular formula of methane?

The first diagram in figure 1 shows two H atoms in the plane of the sheet of paper (solid lines), one hydrogen atom behind the plane of the paper (dotted lines) and one hydrogen atom coming forward out of the plane of the paper (solid triangle). Methane is a hydrocarbon.

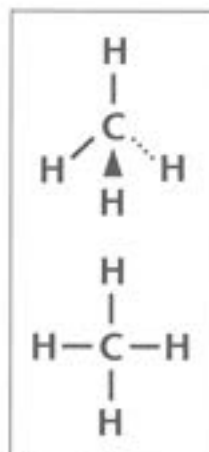


Figure 1

4 Hydrocarbons consist of only two elements. What are these elements?

Alkanes Methane is the first in a series of structurally similar hydrocarbons called **alkanes**. All alkanes have a name ending in '-ane'. The next alkanes after methane are ethane, propane and butane. Their structures are shown in figure 2. Each straight line in figure 2 stands for a single, covalent bond (i.e. a shared pair of electrons). Remember that these are *not* flat, two-dimensional molecules, but three-dimensional (zigzag) molecules.

5 Give the formula for each alkane.

- a ethane b propane c butane

Saturated means 'to hold as much as can be held'. Alkanes are 'saturated' hydrocarbons; that is, all carbon-to-carbon bonds are single bonds and there are no available electrons for 'extra' atoms to bond with. Carbon has a unique ability for **concatenation**; that is, it can form long chains (concatenation means *to link*). The alkanes are a **homologous series**. This means they have similar structures. They differ by a single repeated unit, CH_2 , and show a gradual change in physical properties according to chain length. At room temperature, the first four alkanes are gases, the next twelve are liquids and the remaining alkanes are solids.

6 The next members of the alkanes are pentane, hexane, heptane, octane, nonane and decane. Give the structural and molecular formulae for each of these molecules.

7 Explain what is meant by the statements: 'Alkanes have the general formula of $\text{C}_n\text{H}_{2n+2}$ '

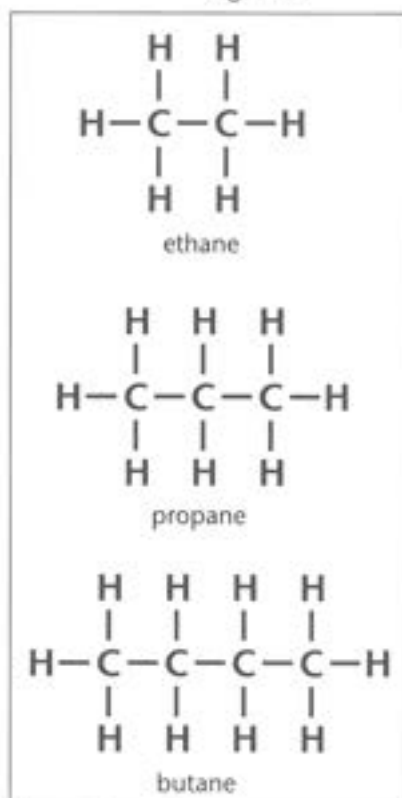


Figure 2

Large deposits of alkanes are quite widespread as fossil fuels (e.g. coal, oil and natural gas). Longer chain alkanes are mainly isolated by the fractional distillation of crude oil (petroleum). Alkanes are relatively unreactive. One major reaction alkanes undergo is combustion.

- 8 Do alkanes conduct electricity? Explain your answer. (Hint: think about bonding.)
- 9 What is another name for *combustion*?
- 10 Given that the products of the combustion of any hydrocarbon (in excess oxygen) are carbon dioxide and water*, write a balanced formula equation for the combustion of butane.
(* Note that if insufficient oxygen is present, hydrocarbons burn to form water and carbon and/or carbon monoxide.)

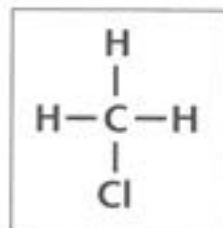


Figure 3

Substituted alkanes The hydrogen in alkanes can be replaced by other elements, especially halogens. Figure 3 shows the structural formula of monochloromethane (CH_3Cl).

- 11 Give the molecular and structural formula of each compound.
 - a dichloromethane
 - b trichloromethane
 - c tetrachloromethane

Alkenes The alkenes are another family of hydrocarbons. They have the general formula C_nH_{2n} . The main differences between alkenes and alkanes are that alkenes have at least one carbon-carbon double bond (two shared pairs of electrons) and they are much more reactive than alkanes.

Alkenes are called 'unsaturated' hydrocarbons. To name an alkene, the ending '-ene' is added to the root derived from the corresponding saturated hydrocarbon.

- 12 How does the general formula for alkenes compare with that of alkanes?
- 13 Suggest why alkenes are more reactive than alkanes.

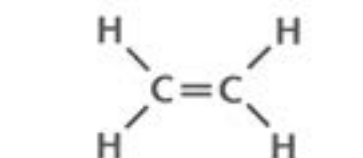
Figure 4 shows the structural formulae of ethene, propene and butene (sometimes called ethylene, propylene and butylene).

- 14 Explain why methene cannot exist.
- 15 Give the molecular formula for each alkene.
 - a ethene
 - b propene
 - c butene

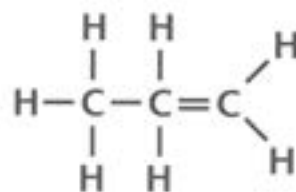
Alkynes Carbon to carbon *triple* bonds can also exist and these bonds are found in alkynes. Alkynes are very reactive hydrocarbons. To name an alkyne, add the ending '-yne' to the root derived from the corresponding saturated hydrocarbon. Figure 5 shows the structural formulae of ethyne and propyne. Like alkanes and alkenes, alkynes also form a homologous series.

Ethyne is the most common alkyne. Its common name is *acetylene*. With oxygen, it burns at a very high temperature and is used for welding metals.

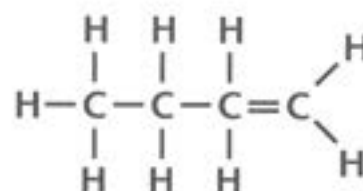
- 16 Suggest why alkynes are more reactive than alkenes.
- 17 Give the molecular formula for each alkyne.
 - a ethyne
 - b propyne
- 18 Give the name, structural formula and molecular formula for the next alkyne in the series after ethyne and propyne.
- 19 Write a balanced equation for the combustion of ethyne.
- 20 Give the general formulae for alkanes, alkenes and alkynes.



ethene



propene

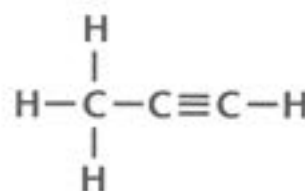


butene

Figure 4



ethyne



propyne

Figure 5

Alkanols

Methanol is the first in a series of structurally similar compounds called **alkanols** (alcohols). All alkanols have a name ending in '-ol'. They are similar in structure to alkanes, but they include a hydroxy (OH) group in the place of one hydrogen atom. Figure 1 shows the structural formula for methanol.

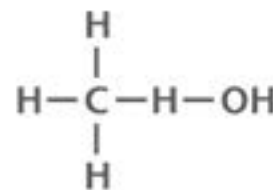


Figure 1

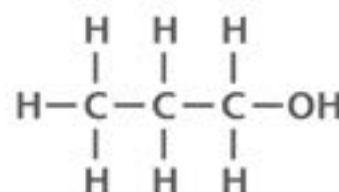
- 1 Give the molecular formula for methanol.

The next alkanols are ethanol, propanol and butanol. The structural formulae of propanol and butanol are shown in figure 2.

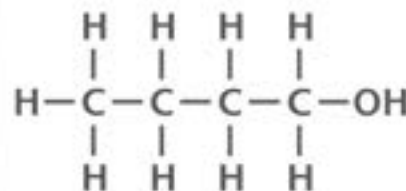
- 2 Ethanol has the formula $\text{CH}_3\text{CH}_2\text{OH}$. Draw the structural formula for ethanol.
- 3 Give the molecular formula for propanol.
- 4 Give the molecular formula for butanol.
- 5 The next alkanol in this homologous series is pentanol. Draw its structural formula and give its molecular formula.

Common alkanols are liquid at room temperature and can be used as solvents, fuels and raw materials (with alkanoic acids) to make esters.

- 6 When ethanol is burned in excess oxygen, carbon dioxide and water are produced. Write a balanced formula equation for this reaction.



propanol



butanol

Figure 2

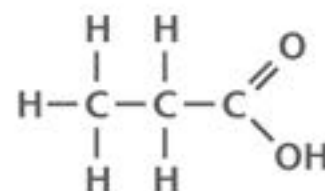
Alkanoic (carboxylic) acids

These organic acids are much weaker than common mineral acids such as sulfuric acid, nitric acid and hydrochloric acid. The first five alkanoic acids are methanoic, ethanoic, propanoic, butanoic and pentanoic acid. They are all liquids. Ethanoic acid is sometimes referred to as acetic acid. In dilute solution, it is vinegar. Figure 3 shows the structural formula for propanoic acid and butanoic acid.

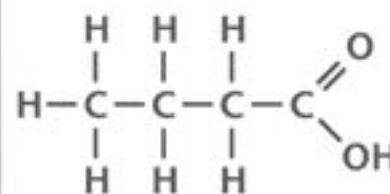
- 7 Draw the structural formula and give the molecular formulae for methanoic acid and ethanoic acid.
- 8 Give the molecular formula for propanoic acid and butanoic acid.

Figure 4 shows the functional groups of alkanols and alkanoic acids. Note that an **alkyl** is a group that is formed when a hydrogen atom is removed from an alkane.

- 9 Explain, with reference to alkanols and alkanoic acids, what is meant by a *functional group*.



propanoic acid



butanoic acid

Figure 3

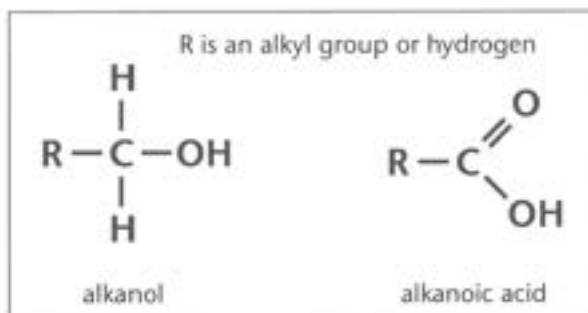


Figure 4

In chemistry, substances can be defined as acidic, basic or neutral. Using an indicator is one way of identifying into which of these categories a substance fits.

- 1 Give your own definition of an indicator.

An **indicator** is a substance that shows the presence of a chemical, usually by changing colour.

Note that sometimes alkaline solutions are mentioned. Remember that a base is the opposite of an acid. If the base is soluble, it can be referred to as an alkali.

The pH scale is often used when referring to acids and bases. A simplified version of it is shown below in figure 1.

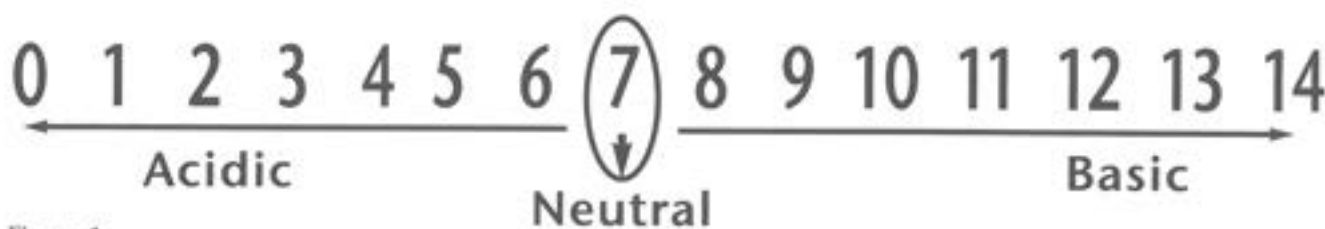


Figure 1

- 2 What pH number coincides with a neutral solution?
- 3 One solution has a pH of 2 and another solution has a pH of 4. They are both acidic. What is the pH of the more acidic solution?
- 4 One solution has a pH of 9 and another solution has a pH of 11. They are both basic. What is the pH of the more basic (alkaline) solution?
- 5 Discuss the statement: 'All alkalis are bases but not all bases are alkalis'.

The earliest indicators were made from natural sources. Some good sources of indicators are blueberries, strawberries, red cabbage, beetroots, cherries and geraniums.

- 6 Name two things that the sources of indicators mentioned above have in common.

Many plants that are red, blue or purple contain a class of compounds (anthocyanins) that change colour with pH. Litmus is extracted from a type of lichen and is one of the most commonly used indicators. Litmus solution has two colours: blue and red. Blue litmus turns red in the presence of an acidic solution. Red litmus turns blue in the presence of a basic solution.

- 7 A solution was added to blue litmus, which didn't change colour. Does this prove the solution is neutral? Explain your answer.
- 8 Synthetic indicators also exist; phenolphthalein is one example. What is meant by *synthetic*?
- 9 What properties would a synthetic indicator have to possess?

It is a common misconception that all indicators change colour at or around pH 7. Indicators can change over a wide range of pHs. Table 1 shows the pH range of some common indicators.

- 10 For each indicator, state the resulting colour when added to a pH 4 solution.

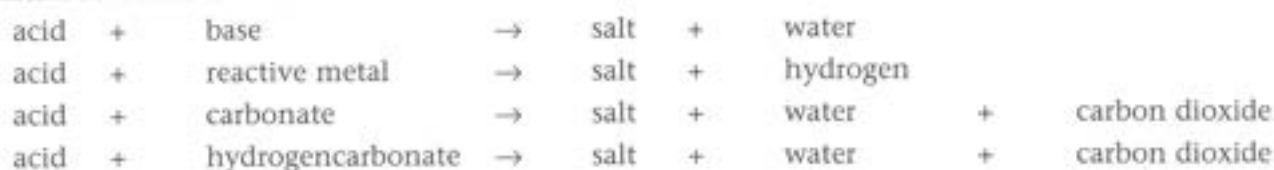
- a litmus
- b phenolphthalein
- c turmeric

- 11 Which of the indicators in table 1 changes colour over the widest pH range?

Indicator	pH range	Colour change
Litmus	5.0 – 8.2	Red to blue
Phenolphthalein	8.2 – 10	Colourless to pink
Turmeric	7.4 – 8.6	Yellow to red
Methyl orange	3.1 – 4.4	Red to yellow

Table 1

The general reactions of common acids can be summarised as follows:



When an acid is neutralised, a salt is produced. This reaction is called a **neutralisation**. Each acid produces a certain type of salt. The salt produced contains most of the elements that were present in the acid from which it was made. Salts do not contain hydrogen ions – these have been replaced by metal ions. Common table salt is called *sodium chloride* and has the formula NaCl.

Hydrochloric acid (HCl) forms salts called chlorides ($-\text{Cl}^-$).

Nitric acid (HNO_3) forms salts called nitrates ($-\text{NO}_3^-$).

Sulfuric acid (H_2SO_4) forms salts called sulfates ($-\text{SO}_4^{2-}$).

Phosphoric acid (H_3PO_4) forms salts called phosphates ($-\text{PO}_4^{3-}$).

Acid + base

You can tell when an acid has been neutralised by a base (or vice versa) by adding an indicator. An indicator is a dye that changes colour to indicate whether a substance is acidic or alkaline (basic).

- 1 Write a word equation and then a balanced equation for each reaction between acid and base.
 - a hydrochloric acid and sodium hydroxide (NaOH)
 - b sulfuric acid and copper II oxide (CuO)
 - c nitric acid and ammonium hydroxide (NH_4OH)
 - d phosphoric acid and potassium hydroxide (KOH)

Acid + metal

When reactive metals are added to acids, hydrogen gas (H_2) and a salt are formed. Hydrogen is an invisible, very flammable, lighter-than-air gas; it goes 'POP!' when placed next to a flame. Unreactive metals such as gold (Au) tend not to react with acids.

- 2 Write a word equation and then a balanced equation for each reaction between acid and metal.
 - a hydrochloric acid and magnesium
 - b sulfuric acid and aluminium
 - c nitric acid and zinc
 - d phosphoric acid and iron (assume iron forms the Fe^{3+} ion)

Acid + carbonate

When acids and carbonates react, carbon dioxide (CO_2) gas is given off and water and a salt are formed. Hydrogencarbonates react in a similar manner. When carbon dioxide gas is bubbled through limewater, the limewater changes from clear to cloudy/milky.

- 3 Write a word equation and then a balanced equation for each reaction between acid and carbonate or hydrogencarbonate.
 - a hydrochloric acid and copper (II) carbonate (CuCO_3)
 - b sulfuric acid and potassium carbonate (K_2CO_3)
 - c nitric acid and sodium hydrogencarbonate (NaHCO_3)
 - d phosphoric acid and potassium hydrogencarbonate (KHCO_3)
- 4 Describe a chemical test for the presence of carbon dioxide.
- 5 'A compound that is formed by the replacement of the hydrogen atom(s) of an acid molecule with a metal or a group acting like a metal.' What does this statement define?

The general equation for the reaction of acids with metals is:



Note that not all metals react with acids. The most common acids used in a school laboratory are hydrochloric acid, nitric acid and sulfuric acid.

- 1 Name two metals that tend to not react with dilute acids.
- 2 Give the formula for each acid.
 - a hydrochloric acid
 - b nitric acid
 - c sulfuric acid

A **salt** is an ionic compound that is formed when an acid is wholly or partly replaced by a metal (or a metal radical). When a salt is formed, the reactive metal replaces the hydrogen in the acid. This hydrogen is given off as a gas.

- 3 Give the name of the salt formed by each acid.
 - a hydrochloric acid
 - b nitric acid
 - c sulfuric acid
- 4 Describe the test used to identify hydrogen gas.

Acids are polar covalent compounds that form ions when they dissolve in water.



H_3O^+ is the hydrated hydrogen ion (sometimes called a hydronium ion): $\text{H}^+ + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$. This ion is what gives a typical acid its acidic properties.

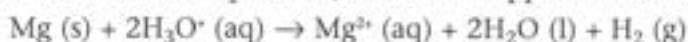
- 5 Name each ion.
 - a Cl^-
 - b SO_4^{2-}
 - c NO_3^-

This reaction shows ions in solution when magnesium reacts with hydrochloric acid:



- 6 A *spectator ion* does not undergo any change in a reaction. Identify the spectator ion that occurs on both sides of the equation.

As with mathematical equations, terms that appear on both sides of the equation can be removed to give:

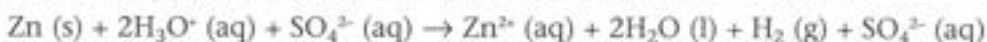


or $\text{Mg (s)} + 2\text{H}^+ \text{ (aq)} \rightarrow \text{Mg}^{2+} \text{ (aq)} + \text{H}_2 \text{ (g)}$

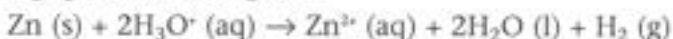
This is called a 'half-equation'.

- 7 When magnesium metal reacts, it changes from Mg to Mg^{2+} . Has it gained or lost electrons?
- 8 When a hydrogen ion reacts, it changes from H^+ to $\frac{1}{2}\text{H}_2$. Note that the convention is to show a hydrogen atom as half of a hydrogen (H_2) molecule. Has the hydrogen ion gained or lost an electron?

This reaction shows ions in solution when zinc reacts with sulfuric acid:



Removing spectator ions gives:



- 9 Identify the ion that occurs on both sides of the equation (i.e. the 'spectator ion').
- 10 When zinc metal reacts, it changes from Zn to Zn^{2+} . Has it gained or lost electrons?
- 11 Write ionic half-equations to show the reactions between each pair of substances.
 - a sulfuric acid and magnesium
 - b hydrochloric acid and zinc

Because individual atoms are so small, units such as the gram are too large for describing low numbers of atoms. Atomic mass units (amu) are the basic units of mass for atoms. One amu is equal to 1.66×10^{-27} kg. Chemists need a standard against which they can compare atoms. The first standard was hydrogen (the simplest atom) which gave a hydrogen atom a mass of 1.0000 amu. Later, naturally-occurring oxygen was used instead of hydrogen and assigned a mass of 16.0000 amu. However, oxygen exists as isotopes, where some atoms have a different number of neutrons and hence different masses. In 1961 the isotope carbon-12 (^{12}C) was adopted as the standard measure and one atom of carbon-12 is defined as having a mass of 12 amu. In other words, the mass of 1 amu is one-twelfth of the mass of a carbon-12 atom. Because the masses of all atoms are compared to this standard, they are often called **relative atomic masses** (*relative* meaning compared to each other). Some older textbooks refer to atomic mass units as **daltons**, named after John Dalton (1766–1844).

Most Periodic Tables give relative atomic masses as well as atomic numbers. Remember:

- **Atomic number** is the number of protons (and is also equal to the number of electrons)
- **Atomic mass (mass number)** is the number of protons plus the number of neutrons
- **Relative atomic masses** of elements are often not whole numbers because in nature they are mixtures of isotopes
- The term **atomic weight** is often used instead of atomic mass.

- 1 Give the term for the definition: 'When a chemical element has atoms with the same atomic number but with differing atomic mass numbers.'
- 2 An atom of germanium is 6 times as heavy as an atom of carbon-12.
 - a Give the relative atomic mass of germanium in amu.
 - b Give the symbol for germanium.
- 3 The relative atomic mass of titanium is 48.
 - a How many times heavier than an atom of carbon-12 is an atom of titanium?
 - b Give the symbol for titanium.

Compounds are groups of atoms joined together. The formula of a compound can be used to calculate its relative formula mass. Sometimes the term **relative molecular mass** is used instead of relative formula mass. Technically, relative molecular mass should only be used when dealing with covalent compounds.

EXAMPLE

What is the mass of a molecule of carbon dioxide CO_2 ?

(Note that the relative atomic mass of carbon is 12.0 and of oxygen is 16.0.)

From the formula, there is one carbon atom and two oxygen atoms in a molecule of carbon dioxide.

$$\text{Mass} = (1 \times 12.0) + (2 \times 16.0) = 12.0 + 32.0 = 44.0 \text{ amu} \quad (\text{Don't forget the units!})$$

- 4 Find the relative formula mass of each compound. Show your working.
 - a magnesium oxide, MgO
 - b iron (III) oxide, Fe_2O_3
 - c copper (II) sulfate, CuSO_4
 - d ammonium chloride, NH_4Cl
 - e sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
 - f potassium carbonate, K_2CO_3
- 5 Mark each statement true or false.
 - a The atomic mass unit (amu) is a heavier unit than the gram.
 - b The atomic mass of an element is the average mass of its isotopes, taking into account the proportion of each isotope in a natural sample.
 - c Relative atomic masses are based on the mass of a naturally occurring hydrogen atom.
 - d One amu is equal to the mass of a neutron.

1 Read the following passage and explain what is meant by each term in *italics*. You may need to use a dictionary or other reference sources.

Alchemy is often defined as the *precursor* of modern chemistry. Alchemy is thought to have *originated* in ancient Egypt, drawing upon the works of some of the early Greek *philosophers*. It also has origins in China where similar ideas arose independently. Alchemy began to influence European thinking and investigations in the 8th century AD when Arabs first introduced it to Spain. Alchemical thinking gradually gained favour during the *Middle Ages* and reached a peak in the 17th century. It was gradually *displaced* by what is now regarded as modern scientific method.

Although alchemists had many reasons for pursuing their researches, alchemy had two major goals. These were to:

- *transmute* base (common) metals into gold
- create a *panacea*.

The philosopher's stone is a *mythical* substance supposed to achieve these goals. Although named a 'stone', it could be in the physical state of a powder or liquid.

Gold was (and still is) a very desirable metal. Alchemists thought that gold was the perfect metal. They believed that if they could find the philosopher's stone (the absolute perfection of gold) then they could use this to change other metals to gold. Gold would become *plentiful* and everyone would become wealthy.

Alchemists believed in the four 'elements' of earth, air, fire and water – a *theory* first advanced by Greek philosophers. All substances were composed of a mixture of these four things. Many alchemists carried out experiments that we would recognise as 'real' chemistry, others had more *dubious* practices. Arabian alchemists had a working knowledge of chemical *reagents* (especially acids, bases and salts) and metals such as mercury and gold. Many discoveries in practical chemistry and *pharmacy* have been *attributed* to researches in alchemy. 'Aqua regia' is a mixture of sulfuric and nitric acids that can dissolve gold. This very corrosive chemical was also thought by some alchemists to be the '*elixir of life*'.

Alchemy can be viewed as a mixture of *astrology*, chemistry, *metallurgy*, medicine, *physics* and religion. Some people have described it as a *proto-science*; others view it as a *pseudo-science*. Alchemy gradually faded away in the 18th century as the work of *systematic* scientists such as Robert Boyle, Antoine Lavoisier and John Dalton gained acceptance. However, even Sir Isaac Newton *devoted* a large amount of his time to the study of alchemy. Tycho Brahe, (the famed *astronomer*) is also reputed to have *dabbled* in alchemy. Although mainly *discredited*, interest in alchemy has survived into the 21st century.

Research

- 2 Outline the alchemical theory that there are four 'elements' of earth, air, fire and water. Include details of the Greek philosophy that this is based on. (Hint: Use the platonic solids in figure 1 to help you.)
- 3 Give another definition for *alchemy*.
- 4 Outline an alchemist's view of the philosopher's stone.
- 5 In the 20th century, physicists managed for the first time to change elements into different ones and make new elements via nuclear reactions. Do you think that this is the modern equivalent of the philosopher's stone? Give reasons for your answer.
- 6 Alchemy continues to appeal to many people. Suggest why, in this day and age, this might be. What is your opinion of alchemy?
- 7 Alchemy was considered a mix of chemistry and magic. Describe specific examples of both magical and chemical applications of alchemy.



Earth



Fire



Air



Water

Figure 1

SELECTED ANSWERS 1–7

1 Separation

1 Particles smaller than the holes in the sieve pass through, larger particles remain behind. 2 Soil, flour 3 The filter paper is folded into a cone (A to D) and then placed in a filter funnel. The mixture is poured into the filter funnel (E). The water passes through the tiny holes in the filter paper leaving the insoluble solid behind (F). 4 Residue is the substance that didn't pass through the filter paper; filtrate is the substance that did pass through the filter paper. 5 G = Bunsen burner, H = hot water, I = evaporating basin, J = solution 6 Water bath has a lower temperature (100°C) than direct heat from Bunsen burner (600–700°C) 7 Resists heat, chemically unreactive, doesn't absorb substances 8 Substance dissolved by a solvent 9 The gas phase of a substance that is usually a liquid at room temperature 10 f, c, a, e, b/d [order interchangeable] 11 a Common salt (sodium chloride) b Sand c 4.8g d 3.5g e Sand = 57.8%, salt = 42.2% f Volume of water g Porous – small holes 12 Particles suspended in a solution, don't pass through filter paper, eventually settle 13 Revolutions per minute, how many times it spins around in a minute 14 Teacher to check 15 Spin drier, automatic washing machine 16 Greater mass per unit volume, particles closer together 17 No, it blends components rather than separating them. 18 Separates out various components 19 Plasma, blood cells

2 The three states of matter

1 Solids have a fixed shape, are usually the most dense shape, have a fixed volume. Liquids have no fixed shape, are almost as dense as their solid phase, are difficult to compress and have a fixed volume. Gases have no fixed shape; take the shape of their container, are much less dense than solids or liquids, have no fixed volume; occupy the volume of their container. 2 Solid – left hand side; liquid – right hand side; gas – top of triangle

3 Separating gases from the atmosphere

1 Steady temperature (under standard conditions) at which a liquid changes to a gas 2 To preserve it; medical samples 3 Not poisonous 4 Cryogenic is using very low temperatures. Superconductivity is the state of very low electrical resistance of a substance at temperatures near absolute zero. 5 Teacher to check

4 Movement of particles

1 Dissolved, solution 2 Particles leave the crystal and spread throughout the liquid. 3 No, they are too small. 4 Colourless 5 Particles, diffusion 6 Less concentrated 7 Particles are in constant, random motion. Students at the front of the class would smell it first, gradually moving to back. 8 Gas particles are further apart, hence can move more freely. 9 True 10 a Stop smoke from escaping b Focus light onto smoke c Magnify smoke particles d To see the smoke particles more easily 11 a Person who studies plants b Not based on carbon c Substance that flows, liquid or gas d Lacking a pattern e Moving f Small piece of something g Substance in gaseous state 12 Diffusion 13 'Deflect' means 'to be redirected', and if the particles are too big, the water molecules won't be big enough to

knock them around. 14 e.g. The random movement of tiny particles in fluid 15 Kinetic

5 Elements

1 Teacher to check 2 Teacher to check 3 Attractive, lustrous, valuable, don't rust (corrode) 4 Don't wear out quickly, can be stamped/embossed 5 Copper, gold 6 The right hand side 7 Semi-metals 8 Copper, iron (steel), silver, etc. 9 False 10 True 11 False 12 True 13 Teacher to check 14 Mercury 15 a Lustrous b Element c Malleable d Magnetic e Ductile f Brittle g Conducts. 16 a Copper (Cu), chlorine (Cl) b Iron (Fe), oxygen (O) c Sodium (Na), chlorine (Cl) d Lead (Pb), iodine (I) e Magnesium (Mg), oxygen (O) f Barium (Ba), sulfur (S) g Calcium (Ca), carbon (C) h Hydrogen (H), bromine (Br) 17 Teacher to check 18 Teacher to check 19 a Calcium (Ca), sulfur (S), oxygen (O) b Copper (Cu), carbon (C), oxygen (O) c Silver (Ag), nitrogen (N), oxygen (O) d Sodium (Na), phosphorus (P), oxygen (O) e Nickel (Ni), nitrogen (N), oxygen (O) f Zinc (Zn), sulphur (S), oxygen (O) g Lead (Pb), carbon (C), oxygen (O)

6 Elements research

Argon Ar, 1894, England, from the Greek 'argos' meaning 'lazy'

Chlorine Cl

Helium He, 1895, from the Greek, 'Helios' meaning 'Sun'

Mercury Hg, ancient, from 'hydrargyrum' or 'liquid silver'

Sulfur S

Uranium U, 1789, Germany, after the planet Uranus

Oxygen O, 1772, Sweden, 'oxygene', meaning 'acid former'

Copper Cu, ancient

Carbon C, ancient, from Latin 'carbo' meaning charcoal

7 The Periodic Table

1 Atomic mass is the quantity of matter in an atom and is expressed in amu. It is approximately equal to the number of protons added to the number of neutrons. Atomic number is the number of protons (= number of electrons) in an atom. 2 I (1), II (2), III (3), IV (4), V (5), VI (6), VII (7), VIII (8) 3 Arranged in order of increasing atomic number (not mass), extra elements 4 Teacher to check 5 Groups are vertical (columns), each group often containing elements with similar properties; Periods go across the Periodic Table (rows), showing elements with varying properties. 6 Between Groups II and III 7 Junction between metals and non-metals 8 Have some metallic and some non-metallic properties 9 Germanium (Ge) and silicon (Si) 10 Name, symbol, atomic number, atomic mass, sometimes electronic configuration 11 Outside electron involved in bonding 12 A metal atom (in the solid state) loses an electron and forms a metal ion with a single positive charge. 13 A metal ion with a single positive charge loses an electron and forms a metal ion with a double positive charge. 14 $\text{Na (s)} + \text{H}_2\text{O (l)} \rightarrow \text{NaOH (aq)} + \frac{1}{2}\text{H}_2 \text{ (g)}$, or $2\text{Na (s)} + 2\text{H}_2\text{O (l)} \rightarrow 2\text{NaOH (aq)} + \text{H}_2 \text{ (g)}$ 15 Li^+ , Na^+ , K^+ , Rb^+ , Cs^+

SELECTED ANSWERS 7–12

and Fr^+ 16 Increases 17 Has 'lost' an outer shell
 18 Be^{2+} , Mg^{2+} , Ca^{2+} , Sr^{2+} , Ba^{2+} and Ra^{2+} 19 Form positive ions, react with water 20 Group 1 form unipositive ion (+1) and Group 2 form dispositive (+2). Group 1 is more reactive than Group 2. 21 Increase as 'extra' electron shells added 22 Both, as it is a metalloid/semi-metal
 23 Non-metal to semi-metal to metallic character
 24 Individual layers, layers internally strongly bonded, weak bonds between layers 25 Delocalised electrons
 26 Allotrope 27 e.g. Diamond: clear, hard, strongly bonded lattice; Graphite: dark, slippery, soft 28 Two atoms joined to make a molecule 29 N_2 (g)
 30 Oxygen 31 Mixture of metals 32 Electrical conductivity is between that of a conductor and that of an insulator 33 H_2SO_4 34 $\text{S}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{SO}_2(\text{g})$
 35 $\text{F}_2(\text{g}) + 2\text{Na}(\text{s}) \rightarrow 2\text{NaF}(\text{s})$ 36 Too chemically reactive 37 $70 \div 1\,000\,000 \times 1 = 0.000\,07\text{ kg} = 0.07\text{ g}$
 38 True 39 Full outer electron shells 40 No valence electrons 41 e.g. Shiny, hard, good thermal and electrical conductor, etc.

42 Titanium Ti	Iron Fe	Zinc Zn
Vanadium V	Cobalt Co	Silver Ag
Chromium Cr	Nickel Ni	Gold Au
Manganese Mn	Copper Cu	Platinum Pt

43 Separate, not joined to others 44 a False b True
 c False d True e False

8 The development of atomic theory 1

1 New idea, couldn't see them 2 e.g. Atoms can be changed in nuclear reactions, isotopes exist 3 a Yes b Yes c Yes d Yes 4 Teacher to check but should be consistent with description 5 Teacher to check but should be consistent with description 6 Element that exists with different numbers of neutrons in nucleus
 7 Examples include: he made major contributions to understanding break down and changing of the radioactive elements, discovered and named particles involved in nuclear radiation, identified the alpha particle as a helium nucleus, evolved the nuclear theory of atomic structure, produced the first artificial disintegration of elements. Many discoveries were made by the people he trained. 8 Any consistent argument, e.g. mention of round particles moving, colliding, etc. 9 Vacuum is the absence of any substance; used so that alpha particles wouldn't collide with anything 10 Most of particles went straight on because they did not meet anything substantial (i.e. the nucleus). Those that came close would be deflected and those that hit nucleus would bounce back. Nucleus couldn't be very large or most of the particles would be deflected. 11 Lead blocks most radiation to ensure radiation only went towards target
 12 Properties of nucleus, existence of electrons, electrons rotating around nucleus 13 Electrons associated with specific 'orbits' 14 Probability of finding an electron increases to a maximum at a certain point and then decreases 15 Bohr's model is more definite about position of electron.

9 The development of atomic theory 2

1 Particles that make up the atom

	Proton	Neutron	Electron
Position	nucleus	nucleus	orbiting
Charge	plus	neutral	minus
Mass (amu)	1	1	0.0005

3 a 54 b 25 c 25 d 25 e 29 4 63.62 amu
 5 Named after John Dalton 6 a Carbon; 6; 2, 4
 b Fluorine; 9; 2, 7 c Neon; 10; 2, 8 d Sodium; 11; 2, 8, 1 e Sulfur; 16; 2, 8, 6 f Calcium; 20; 2, 8, 8, 2
 7 Eight 8 Neon, argon, krypton, xenon and radon (any 3) 9 Group VIII (0) 10 Full outer electron shells
 11 Bonding/outer electrons in incomplete shells
 12 There will be one more (positive) proton than (negative) electrons.

10 Forming ions

1 3+ 2 Teacher to check diagrams, Na^+ , Mg^{2+} , Be^{2+} , Al^{3+} , K^+
 3 Metals, lose electrons, form positive ions 4 1–
 5 Teacher to check diagrams, O^{2-} , Cl^- , S^{2-} , Br^- 6 Non-metals, initially have incomplete outer shells, gain electrons, form negative ions 7 a Lithium (I)
 b Sodium (I) c Magnesium (II) d Beryllium (II)
 e Aluminium (III) f Potassium (I) g Fluorine (VII)
 h Oxygen (VI) i Chlorine (VII) j Sulfur (VI) k Bromine (VII) 8 Groups I to III tend to lose, Groups VI to VII tend to gain; e.g. Group I elements tend to lose one electron, Group II elements lose two electrons, and so on 9 Yes – transfer of electrons from metal to non-metal – ionic bond

11 Lewis electron dot structures

1 Negative 2 They are 'full', not involved in bonding
 3 How electrons are arranged in shells 4 Sodium atom transfers an electron to a chlorine atom, forming a (positive) sodium ion and a (negative) chlorine ion, giving each atom a complete outer shell. 5 K, 2, 8, 8, 1
 6 F, 2, 7 7 Teacher to check: forming K^+ and F^- 8 Ionic
 9 Teacher to check: forming Mg^{2+} and O^{2-} 10 Chlorine atoms share a pair of electrons to form a covalent bond
 11 Teacher to check 12 Oxygen atoms share 2 pairs of electrons forming 2 covalent bonds 13 Ionic: transfer, covalent: sharing 14 See figure 1. 15 Acids are electron acceptors and bases are electron donors.



Figure 1

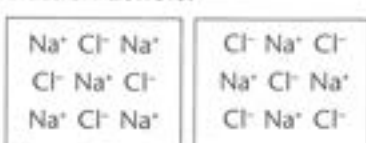


Figure 2

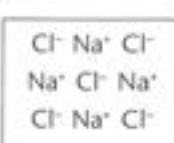


Figure 3

12 Ionic and covalent compounds

1 NaF , MgCl_2 , CaI_2 , LiBr 2 See figure 2. 3 See figure 3.
 4 Actual numbers will vary, ratio will stay the same
 5 Negative 6 Positive 7 Full outer electron shell
 8 Covalent, metallic 9 Noble gases 10 Strong attractive forces between ions 11 Teacher to check, show positive ions moving towards negative electrode, and negative

SELECTED ANSWERS 12–16

ions towards positive electrode 12 Ions not mobile when solid 13 a Ionic b Cation c Valence d Electrolysis e Empirical f Solvent g Octet h Polar 14 CCl_4 15 a No b Covalent – between non-metals 16 Weak intermolecular bonds 17 Teacher to check; should discuss how diamond fits the description of a macromolecule 18 CH_4 19 Low, covalent, discrete molecules with weak forces of attraction between them 20 Covalent – change to gases easily, hence weak forces between molecules 21 e.g. Ethanol ($\text{C}_2\text{H}_5\text{OH}$) and tetrachloromethane (CCl_4) 22 No ions or free electrons 23 'Like dissolves like.' 24 Carbon-based 25 NH_3 26 See figure 4. 27 Polar, water is a polar substance 28 Teacher to check; should include reference to melting and boiling points, solubility in water, ions versus molecules

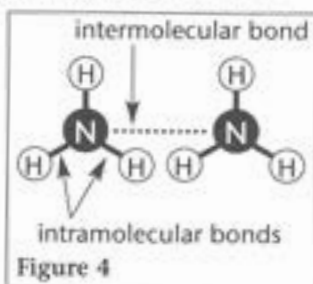


Figure 4

13 Metals and metallic bonding

1 e.g. Shiny, malleable, ductile, hard, good thermal and electrical conductor, etc. 2 Positive, or cations 3 Full outer electron shells 4 Outer, bonding electrons 5 Heat 6 Can be drawn into a wire 7 Have delocalised electrons 8 As temperature increases, electrical conductivity decreases 9 a Metallic bonding b Alloy c Cation d Malleable e Lattice 10 Metallic is with other metal atoms and have delocalised electrons, ionic is with non-metals and electrons transferred to non-metals.

14 Chemical reactions

1 Condensing 2 No new substance is formed, substance is changing appearance 3 Teacher to check 4 a Physical b Physical c Chemical d Physical e Chemical f Physical g Chemical 5 Teacher to check 6 e.g. Can't be reversed, new substance formed 7 Electrical 8 Heat and light 9 Teacher to check; e.g. explosion 10 Teacher to check; e.g. iron rusting 11 Energy needed to start chemical reaction 12 a e.g. Wax combines with oxygen to form carbon dioxide and water, giving out heat and light b Wax melts 13 Bonds are broken and new ones formed 14 a The starting materials b Materials made by the reaction 15 a Products go from a minimum to a maximum and then amount stays constant b Reactants go from a maximum to a minimum and then amount stays constant 16 a Heat given out to surroundings b Heat taken in from surroundings 17 a Mercury oxide forms mercury and oxygen – decomposition b Hydrogen and bromine form hydrogen bromide – synthesis c Methane and oxygen form carbon dioxide and water – combustion d Nitrogen and hydrogen form ammonia – synthesis e Calcium carbonate forms calcium oxide and carbon dioxide – decomposition 18 Teacher to check 19 True

15 Naming compounds

1 Non-metals, when in a compound ending of name changes to '-ide' 2 Copper chloride; copper and chlorine; Cu, Cl

Zinc oxide; zinc and oxygen; Zn, O
Iron sulfide; iron and sulfur; Fe, S
Sodium iodide; sodium and iodine; Na, I
Magnesium nitride; magnesium and nitrogen; Mg, N
Potassium bromide; potassium and bromine; K, Br
3 CuSO_4 ; copper, sulfur, oxygen; copper sulfate
 FePO_4 ; iron, phosphorus, oxygen; iron sulfate
 AgNO_3 ; silver, nitrogen, oxygen; silver nitrate
 CuCO_3 ; copper, carbon, oxygen; copper carbonate
 BaSO_4 ; barium, sulfur, oxygen; barium sulfate
 NaNO_3 ; sodium, nitrogen, oxygen; sodium nitrate
4 Hydrogen and oxygen 5 a Nitrogen and hydrogen b Hydrogen and oxygen c Carbon, hydrogen and oxygen 6 a Manganese oxide, manganese (1), oxygen (2) b Cobalt chloride, cobalt (1), chlorine (2) c Magnesium sulfate, magnesium (1), sulfur (1), oxygen (4) d Aluminium chloride, aluminium (1), chlorine (3) e Zinc hydroxide, zinc (1), oxygen (2), hydrogen (2) f Copper carbonate, copper (1) carbon (1), oxygen (3) g Aluminium oxide, aluminium (2), oxygen (3) h Barium nitrate, barium (1), nitrogen (2), oxygen (6) i Potassium phosphate, potassium (3), phosphorus (1), oxygen (4) 7 Don't form ions/bonds 8 a PCl_3 , phosphorus (tri) chloride b CO_2 , carbon dioxide c B_2O_3 , boron oxide d CS_2 , carbon disulfide e NH_3 , nitrogen hydride (ammonia) f H_2S , hydrogen sulfide g SiF_4 , silicon (tetra) fluoride 9 Radicals 10 a True b True c False d False e True f True 11 a 1 calcium, 2 oxygen, 2 hydrogen b 3 calcium, 2 phosphorus, 8 oxygen c 1 zinc, 2 nitrogen, 6 oxygen d 2 sodium, 1 carbon, 3 oxygen e 1 potassium, 1 fluorine f 1 copper, 1 sulfur, 4 oxygen g 2 nitrogen, 8 hydrogen, 1 sulfur, 4 oxygen 12 a +1 b +2 c +3

13 (In table order)

Sodium, Na^+ : NaCl , Na_2O , Na_2S , Na_3N
Magnesium, Mg^{2+} : MgCl_2 , MgO , MgS , Mg_3N_2
Iron (III), Fe^{3+} : FeCl_3 , Fe_2O_3 , Fe_2S_3 , FeN
Ammonium, NH_4^+ : NH_4Cl , $(\text{NH}_4)_2\text{O}$, $(\text{NH}_4)_2\text{S}$, $(\text{NH}_4)_3\text{N}$
Lithium, Li^+ : LiOH , Li_2SO_4 , Li_2CO_3 , Li_3PO_4
Calcium, Ca^{2+} : $\text{Ca}(\text{OH})_2$, CaSO_4 , CaCO_3 , $\text{Ca}_3(\text{PO}_4)_2$
Aluminium, Al^{3+} : $\text{Al}(\text{OH})_3$, $\text{Al}_2(\text{SO}_4)_3$, $\text{Al}_2(\text{CO}_3)_3$, AlPO_4
Ammonium, NH_4^+ : NH_4OH , $(\text{NH}_4)_2\text{SO}_4$, $(\text{NH}_4)_2\text{CO}_3$, $(\text{NH}_4)_3\text{PO}_4$

14 Valence of 2 for iron 15 Transition metals
16 Valency

16 Valency and formula

1 Teacher to check; e.g. property of an element governing the number of other atoms with which an atom of the element can combine 2 Teacher to check; e.g. Groups are vertical columns, Periods horizontal rows 3 2, gain two electrons to fill outer shell 4 Teacher to check 5 Metals, positive ions 6 NH_4^+ 7 +1 8 Cu^+ , Cu^{2+} 9 Teacher to check; e.g. elements that have atoms with valence electrons in more than one shell 10 e.g. Iron, copper, zinc 11 Transition – passing from one position or state to another. They are the link, or transition, from metal to non-metal in the Periodic Table. 12 Teacher to check 13 Non-metals, negative ions 14 a HPO_4^{2-} b HCO_3^- c SO_4^{2-} d CrO_4^{2-}

SELECTED ANSWERS 16–23

15 a Carbon (2), oxygen (4) b Phosphorus (1), oxygen (4) c Carbon (1), oxygen (3) 16 Charged group of atoms in compounds that always exists as a group
17 Teacher to check 18 a Positive b Negative
19 Teacher to check 20 a NaOH b Ca(OH)₂ c MgO
d CuCl₂ e CuCl₂ f (NH₄)₂CO₃ g Fe₂O₃ h Ba(NO₃)₂
i CuSO₄ j Ni₃(PO₄)₂ k PbS₂ l K₂CrO₄ m Ba₃P₂ n LiCN o Al₂O₃

21

Bicarbonate of soda; sodium hydrogencarbonate; NaHCO₃
Slaked lime; calcium hydroxide; Ca(OH)₂
Gypsum; calcium sulfate; CaSO₄
Epsom salts; magnesium sulfate; MgSO₄
Condy's crystals; potassium permanganate; KMnO₄
Glauber's salt; sodium sulfate; Na₂SO₄
Caustic soda; sodium hydroxide; NaOH
Saltpetre; potassium nitrate; KNO₃
Potash; potassium carbonate; K₂CO₃

22 a Magnesium, Mg²⁺; sulfate, SO₄²⁻ b Potassium, K⁺; nitrate, NO₃⁻ 23 a False b False c True d False
e True f True g False h False

17 Introduction to carbon chemistry I

1 CO₂ 2 Sharing 3 CH₄ 4 Carbon and hydrogen
5 a C₂H₆ b C₃H₈ c C₄H₁₀ 6 C₅H₁₂, C₆H₁₄, C₇H₁₆, C₈H₁₈, C₉H₂₀, C₁₀H₂₂ 7 For every carbon atom, there are twice as many hydrogen atoms, plus 2 more 8 No, covalent – no ions or free electrons 9 Burning
10 C₄H₁₀ (g) + 6½ O₂ (g) → 4CO₂ (g) + 5H₂O (l)
11 a CH₂Cl₂ b CHCl₃ c CCl₄ 12 Similar but has two less hydrogen atoms per molecule 13 Carbon-carbon double bond 14 Only one carbon atom present
15 a C₂H₄ b C₃H₆ c C₄H₈ 16 Carbon-carbon triple bond 17 a C₂H₂ b C₃H₄ 18 Butyne, C₄H₆
19 C₂H₂ (g) + 2½ O₂ (g) → 2CO₂ (g) + H₂O (l)
20 C_nH_{2n+2}, C_nH_{2n}, C_nH_{2n-2}

18 Introduction to carbon chemistry 2

1 CH₄O or CH₃OH 2 CH₃CH₂OH, teacher to check
3 C₃H₈O or C₃H₇OH 4 C₄H₁₀O or C₄H₉OH
5 CH₃CH₂CH₂CH₂CH₂OH, C₅H₁₂O or C₅H₁₁OH; teacher to check
6 C₂H₅OH (l) + 3O₂ (g) → 2CO₂ (g) + 3H₂O (l)
7 HCOOH, CH₂O₂; CH₃COOH, C₂H₄O₂; teacher to check
8 C₃H₆O₂, C₄H₈O₂ 9 Group of atoms attached to carbon chain that gives compounds their specific properties, i.e. -OH in alkanols and -COOH in alkanolic acids

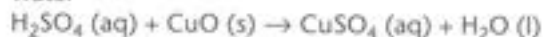
19 Acids, bases and indicators

1 Teacher to check; e.g. substance that changes colour to tell if a substance is acidic or basic 2 7 3 pH 2 4 pH 11
5 Teacher to check 6 Coloured, plants 7 No, could have been basic 8 Made by humans 9 Teacher to check; e.g. similar to those of natural 10 a Red
b Colourless c Yellow 11 Litmus

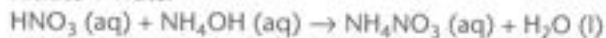
20 Acid reactions

1 a Hydrochloric acid + sodium hydroxide → sodium chloride + water
HCl (aq) + NaOH (aq) → NaCl (aq) + H₂O (l)

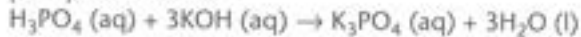
b Sulfuric acid + copper (II) oxide → copper sulfate + water



c Nitric acid + ammonium hydroxide → ammonium nitrate + water



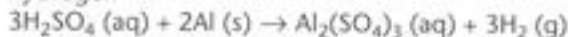
d Phosphoric acid + potassium hydroxide → potassium phosphate + water



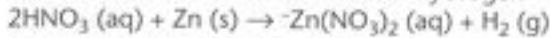
2 a Hydrochloric acid + magnesium → magnesium chloride + hydrogen



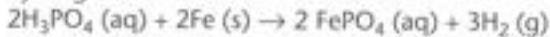
b Sulfuric acid + aluminium → aluminium sulfate + hydrogen



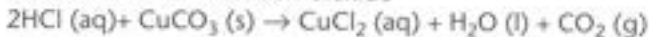
c Nitric acid + zinc → zinc nitrate + hydrogen



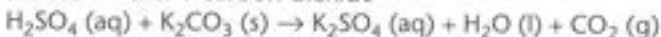
d Phosphoric acid + iron → Iron (III) phosphate + hydrogen



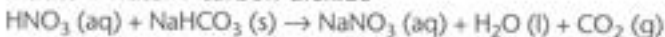
3 a Hydrochloric acid + copper carbonate → copper chloride + water + carbon dioxide



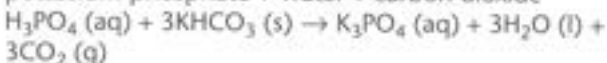
b Sulfuric acid + potassium carbonate → potassium sulfate + water + carbon dioxide



c Nitric acid + sodium hydrogencarbonate → sodium nitrate + water + carbon dioxide



d Phosphoric acid + potassium hydrogencarbonate → potassium phosphate + water + carbon dioxide



4 Limewater turns from clear to cloudy/milky if carbon dioxide is bubbled through it. 5 Salt

21 Acids and metals

1 Gold, silver, platinum, etc. 2 a HCl b HNO₃
c H₂SO₄ 3 a Chlorides b Nitrates c Sulfates
4 Burns with a loud 'pop' when placed in contact with a naked flame 5 a Chloride b Sulfate c Nitrate
6 Chloride (Cl⁻) 7 Lost 8 Gained 9 Sulfate (SO₄²⁻)
10 Lost 11 a 2H⁺ (aq) + Mg (s) → H₂ (g) + Mg²⁺ (aq)
b 2H⁺ (aq) + Zn (s) → H₂ (g) + Zn²⁺ (aq)

22 Relative atomic mass

1 Isotope 2 a 72 amu b Ge 3 a 4 b Ti 4 a 24.31 + 16.00 = 40.31 amu b 2 (55.85) + 3 (16.00) = 159.70 amu
c 63.55 + 32.06 + 4 (16.00) = 159.60 amu
d 14.01 + 4 (1.01) + 35.45 + 53.5 amu e 12 (12.01) + 22 (1.01) + 11 (16.00) = 342.34 amu f 2 (39.10) + 12.01 + 3 (16.00) = 138.21 amu 5 a False b True
c False d False

23 Alchemy: the ancestor of chemistry

Teacher to check

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Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Assignments in Chemistry Book 1

This new three-book chemistry series by highly-experienced science authors and teachers Peter Gribben and Wendy St Germain provides photocopiable worksheets designed to challenge and stimulate students, as they explore a variety of chemistry concepts.

There are an array of opportunities for students to test their scientific skills through the analysis and interpretation of diagrams, tables and graphs. Students will also extend their reasoning and literacy skills as they are guided through the application of theoretical concepts to practical examples.

Research tasks are included to expand students' knowledge of the history of chemistry and the contributions of individual scientists.

Each assignment is self-contained and requires minimal teacher preparation. Answers have been provided for ease of marking.

Topics include

Separating substances
The three states of matter
Movement of particles
Elements
The Periodic Table
Development of atomic theory
Ionic and covalent compounds

Metals and metallic bonding
Valency and formula
Carbon chemistry
Indicators
Acid reactions
Relative atomic mass

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